SUSTAINABLE CHARCOAL AND IMPROVED COOKSTOVE INITIATIVE USING MICRO-GASIFIER IN INDIA

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Project Title	Sustainable charcoal and Improved cookstove initiative using Micro-Gasifier in India
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Summary description of the technologies/measures to be implemented by the project

The proposed project aims at reducing the fuel wood consumption, in traditional cook stoves by disseminating TLUD micro-gasifiers stoves at subsidized prices. Moreover, the project will facilitate the collection of charcoal generated in micro-gasifiers stoves from stove users and the sale of this charcoal to eligible demand.

Overall objectives are reduction of greenhouse gases, conservation of forests and woodlands as well as improved health conditions of micro-gasifier users due to improvement in indoor air quality.

The scenario existing prior to the implementation of the project activity instances is the usage of traditional wood fuel and inefficient cookstoves with poor ventilation, causing excessive indoor air pollution (IAP) and has been a serious health risk for women and children who spend several hours in the kitchen for cooking as well as non-cooking purposes. And the consumption of inefficient and non-renewable conventional charcoal by traditional charcoal users.

The end users of micro-gasifier stoves and charcoal will be informed in advance that the use of micro-gasifier will generate carbon credits which in turn is used for subsiding the price of the micro-gasifiers and for recovering the implementation costs. And in long run, the project plan to procure renewable biomass source and provide them to some of the micro-gasifier users.

ECO-NEXUS Carbon Credit (ENCC) and Korea Center for Carbon Trade Standards (KCCTS) are coming together for the implementation of the project. ENCC will be co-investor and manage the whole project as project proponent, KCCTS participate as co-investor of the project and support ENCC for carbon financing and project development. Other entities may also participate as a co-investor to distribute TLUD and participates as purchaser of carbon credits.

Location of project

The project activity is projected to implement 550,000 TLUD micro-gasifier stoves in the multiple states of India with an initial distribution of 2,000 TLUD micro-gasifier stoves in two districts named Howrah and Purba Medinipur of West Bengal.

An explanation of how the project is expected to generate GHG emissions reductions

The use of micro gasifier will substitute the current common cooking in open fire. The project equipment burns wood more efficiently and hence saving the fuelwood and lowering the GHG emissions. Moreover, the consumption of conventional charcoal by traditional charcoal users shall be reduced by providing them charcoal generated in micro wood gasifier stoves.

An estimate of annual average and total GHG emissions reductions and removals

In the grouped project, the first instance has been started with distribution of 2,000 micro-gasifiers in West Bengal. The grouped project has started distributing of micro-gasifier from 10-September-2022 - and looking forward to distributing more. The estimated total and annual emission reductions for this crediting period of 7 years are **21,393,963 tCO_2e and 3,056,280** tCO_2e/year.

For verification period for the Batch 1 in first instance of 4 months starting from 10-September-2022 till 31-December-2022, the total number of micro- gasifiers which were distributed were 2000 distributed from 10-September-2022 till 25-September-2022, in state of West Bengal.

1.2 Sectoral Scope and Project Type

The project activity under consideration is grouped project.

Sectoral Scope: 03 Energy Demand and 05 Chemical industries

Project Type: Type II Energy Efficiency

Project Type: Type III Others

1.3 Project Eligibility

The grouped project activity includes the distribution of TLUD micro-gasifier units for production of sustainable charcoal and using the heat from charcoal production as thermal energy in cookstove. The cookstove is a single pot improved cooking devices at the household level with the minimum efficiency of 28% which replaces three-stone open cooking methods used in the baseline.

Non-renewable biomass has been used in the project region since 31 December 1989, and wood fuel has been widely used as a fuel for cooking in the baseline, it has been confirmed from the literature[1]. The project activity is a voluntary initiative of the project implementer and meets all the requirements of VCS Standard 4.4.

Methodology

Project activity supported by two methodologies approved by the VCS program activity include

- **VMR0006:** Methodology for Installation of High Efficiency Firewood Cookstoves version 1.1 and

[1] Forest Survey of India Report (fsi.nic.in)

- **AMS-III.BG:** Emission Reduction through sustainable charcoal production and consumption version 04.0

Hence, the project is eligible under the scope of the VCS program.

1.4 Project Design

This project is grouped project under which different instances would be developed. In this grouped project, each TLUD constitutes a single project activity instance which is in line with the definition stated in VCS Program Definitions version 4.3.

Eligibility Criteria

A set of eligibility criteria shall ensure that new project activity instances:

Number	Criterion	How the new project activity instances to comply
1	Meet the applicability conditions set out in the methodology applied to the project	New Project activities instances (micro-gasifiers stove) will meet the applicability conditions set out in section 3.2 Details of applicability conditions for initial project activity instances: Instance 1: initial project activity instances meet respective applicability conditions of methodology, In all other instances the eligibility criteria would be fulfilled.
2	Use of technologies or measures specified in the project description	TLUD (Top-Lit Updraft) Micro-gasifier stoves will be used in the project
3	Apply the technologies or measures in the same manner as specified in the project description	Micro-gasifier stoves and other ICS to be adopted in the project and it will replace the traditional three stone cookstoves in the household
4	Are subject to baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances (each micro-gasifier or ICS) will be distributed within India only and subject to same baseline scenario as determined in section 3.4
5	Have characteristics with respect to additionality that are consistent with the initial instances for the	All new project instances will use the project method for demonstration of additionality. ***Step 1: Regulatory Surplus***

	specified project activity and geographic areas.	There is no mandated government programme or policy in host country of this project ensuring the distribution of new project activity instances. ***Step 2: Simple Cost Analysis*** Where a small subsidy is taken for the purchase of the cookstove, simple cost analysis is carried out. Where a sustainable charcoal value chain is operated based on the premise of carbon financing, simple cost analysis is carried out.

Inclusion of New Project Activity Instances

The inclusion of new project activity instances each of the micro-gasifier cookstoves are included and the project proponent shall ensure that it meets the eligibility criteria below:

Number	Criterion	How the new project activity instances to comply
1	Occur within one of the designated geographic areas specified in the project description	All the instances of the grouped project would be in the project boundary i.e., India. The instances would be implemented within the project boundary.
2	Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial Conformance with multiple sets of eligibility criteria is insufficient.	All the instances of the grouped project will meet the eligibility criteria as set in the section 1.4 of the document.
3	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by validation/verification body.	All the instances would involve all the related information in the monitoring reports including the technical, financial, geographical and other relevant information.
4	Be included in an updated project description, with updated project location information (as set out in Section 3.11 of VCS standard version 4.4), which shall be validated at the	All the instances will include the updated project description and the implementation status of the instance in the PD.

	time of verification against the applicable set of eligibility criteria.	
5	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions)	All the instances involve the distribution of the micro-gasifiers a improved cookstoves and the ownership details of all the instances would be described in the PD
6	Have a start date that is the same as or later than the grouped project start date.	All the instances in this grouped project have the start date same or later than the grouped project start date. The first instance of the grouped project is started from the 10 September 2022 i.e. the start date of the group project.
7	Be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period (only).	All the instances in this grouped project will be in the crediting period.
8	Only eligible for crediting from the start of the verification period in which they were added to the grouped project	The instances will be eligible for the crediting from the start date of implementation.
9	Not be or have been enrolled in another VCS project	The instance in the grouped project would not be registered as the project activity in under VCS, and the project devices have a unique ID associated with them, which would not be double counted in any of the instances or project activity.
10	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9 of VCS standard version 4.4 3.6.8 The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project	Each of the micro-gasifier cookstoves consist single instances, and capacity per 1 unit of TLUD is far smaller than the applicable limit, for both cookstove component applied VMR0006 ver 1.1 and sustainable charcoal component applied

	activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other). 3.6.9 Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified. 2) Such instances shall be divided into clusters, whereby each cluster is comprised any system of such instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	Limits refer to CDM TOOL19 Methodological tool: Demonstration of additionality of microscale project activities Version 10.0 Cookstove component - Type II: Energy efficiency improvement project activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 20 GWh per year (equivalent to 60GWhth per year). For consideration of capacity limit, each project activity instance that does not exceeds one percent of the capacity limit (600 MWh_{th} per year) does not have to be identified As per manufacturer certificate, capacity per TLUD is 1.516 kW_{th} with stove efficiency 28% based on MSME WBT Certificate. Assuming 10% baseline stove efficiency, the energy saving per TLUD shall be 2.808 kW_{th}. Even if the TLUD is operational all year (8760hr/yr), the output will be around 24.598 MW_{th}, which is much smaller than 1% of threshold. Charcoal component - Type III: Other project activities not included in Type I or Type II that result in GHG emission reductions not exceeding 20 kt CO_2e per year in any year of the crediting period. 1% of the capacity limit refers to 200 tCO_2e per year. Even if assuming extreme conditions for example 100% f_{NRB} and CF 6 applied for AMS.3.BG methodology, 1 ton of charcoal refers to around 7.64 tCO_2e which means more than 26 ton /yr

		charcoal production. Reference value of charcoal production of 1-unit TLUD is around 0.18 ton which is much smaller than 26-ton charcoal/yr.

1.5 Project Proponent

Organization name	ECO-NEXUS Carbon Credit Co. Ltd.
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Title	CEO
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Telephone	+82-10-8881-2538
Email	hychung@eco-nexus.co.kr

1.6 Other Entities Involved in the Project

Organization name	**Infinite Environmental Solutions Limited.**
Role in the project	Carbon Project consultant
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Title	COO
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Email	jimmy@infisolutions.org

Organization name	**Sapient InfoTech**
Role in the project	**Project Management**
Contact person	Mr. Moulindu Banerjee
Title	Founder

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1.7 Ownership

The entitlement of micro gasifiers distributed and installed under this project activity is owned by ECONEXUS Carbon Credit Co. Ltd. and distributed by Sapient Infotech as a voluntary contribution towards climate change mitigation and community development.

There is a declaration is signed by the beneficiary of the cookstove to transfer the carbon ownership to the ECONEXUS Carbon Credit Co. Ltd. in exchange of the subsidised micro-gasifier which is distributed and maintained by the Sapient Infotech.

Sapient Infotech also has an agreement with the charcoal users i.e. SMEs in the areas, for the transfer of the carbon ownership to Sapient Infotech and ECONEXUS Carbon Credit Co. Ltd.

Sapient and ECONEXUS has a agreement signed under which all the carbon credit generated from the activity is owned by the ECONEXUS Carbon Credit Co. Ltd. and Sapient would be distributing and maintaining the micro-gasifiers on the field.

1.8 Project Start Date

This is grouped project, into which the Instance 1 has been started, batch wise distribution of the micro-gasifiers and improved cookstoves would be done.

The first micro-gasifier batch has been distributed on 10-September-2022. Thus, the project starts date is 10-September-2022.

1.9 Project Crediting Period

Project Crediting Period: First

Start Date: 10-September-2022 (First operating day of the project activity ICS).

End date: 09-September-2029

Total Number of Years: 7 years

The crediting period shall be renewed twice.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The grouped project activity instances being included currently have more than 300,000 tCO_2e Emission reductions, hence these project activity instances are classified as "Large Projects"

This is developed as a grouped project, and each instance may vary the scale. For the Instance 1 of the project is a Large scale project.

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO_2e)
10-September-2022 – 31 December 2022	3,951
1-January-2023 – 31 December 2023	1,643,025
1-January-2024 – 31 December 2024	3,574,374
1-January-2025 – 31 December 2025	3,530,452
1-January-2026 – 31 December 2026	3,486,970
1-January-2027 – 31 December 2027	3,443,922
1-January-2028 – 31 December 2028	3,401,304
1-January-2029 – 9 September 2029	2,309,366
Total estimated ERs	**21,393,963**
Total number of crediting years	07
Average annual ERs	**3,056,280**

1.11 Description of the Project Activity

The grouped project activity aims at reducing the fuel wood consumption and indoor air pollution due to usage of traditional stove by households in India by disseminating TLUD gasifier at subsidized prices. Where micro gasifier cook stoves will be distributed to the households manufactured in India by Project Management Partner Sapient InfoTech.

Figure 1 Top-Lit Up-Draft Micro Gasifier Units

In the Top-Lit Up-Draft (TLUD) technology dry biomass i.e wood fuel is ignited on the top of a vertical container (canister), starting the process of pyrolysis. During the pyrolysis reaction, combustible gases are generated and rise upwards. The gas is incinerated and used as fuel for cooking. The TLUD allows for significant savings of firewood compared to traditional 3 stone stoves and additionally generates charcoal as by product. TLUD reduces women's time spent on domestic work such as cooking and wood collection.

The TLUD micro-gasifier distributed for this project activity will have an average lifespan of 7 years with a thermal efficiency of about 28%. The project will replace three Stone cookstoves used in the baseline with Improved and fuel-efficient micro gasifier, which will not only help in cooking the food in safe environment but will also produce sustainable charcoal. Most participant communities are poor families and highly dependent on the forest to fulfil their resource demand. This combination of forest fragmentation and fuelwood demand in the area drives the high reliance on fuelwood as the primary source for cooking fuel.

Also, Charcoal generated in the TLUD will be collected from TLUD users and sold to users of conventional charcoal or to local retailers which will sell to the afore mentioned and other SME. No charcoal will be sold to large-scale industries. The consumption of conventional charcoal by selected users of conventional charcoal may be reduced by providing them charcoal generated in the wood gasifier stoves as a by-product. TLUD users may also be provided with renewable biomass to operate ICS, allowing to reduce CO_2e emissions from fuel consumption to zero.

The project aims to reach marginal people with clean cooking solution access in form of TLUD, and also generate clean charcoal using the micro-gasifier. Project implementer train people appropriately at the local level to cater for the households with stove maintenance and basic repairing. After the installation, representatives from the implementing entity performed paperwork and the stove is recorded in the database. A baseline survey was conducted and found that the people were using Three-Stone cookstoves in the baseline for cooking.

1.12 Project Location

Grouped project activity would be in multiple states of India and distributed in phases. Thus, the geographical area of project activity is considered as India.

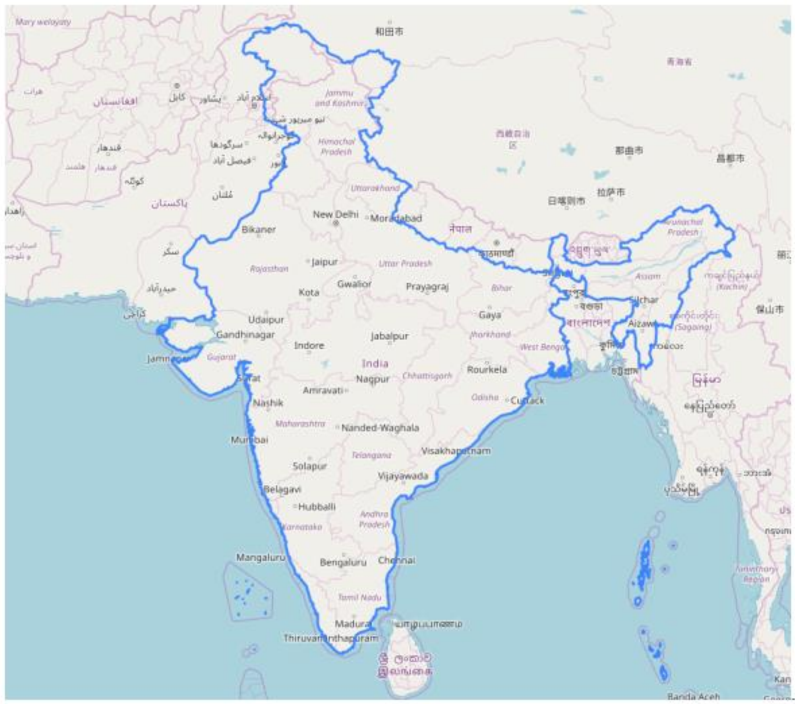


Figure 2 Project Location: India

In the initial phase of the project, the micro gasifier stove distribution has been distributed in the Howrah and Purba Medinipur district in state of West Bengal, India. Distribution details of the instance is described in the table below:

Instance	Batch	Number	Distributing region
First Phase	Batch 1	2000	Howrah and Purba Medinipur, West Bengal

Project Coordinates

	Start Coordinate	End-Coordinate
Latitude	21°2’ N	26°87’ N
Longitude	74°02’ E	82°48’ E

Project aims to distribute initial in the West Bengal, eastern state near to Bangladesh.

Figure 3 Location of West Bengal.

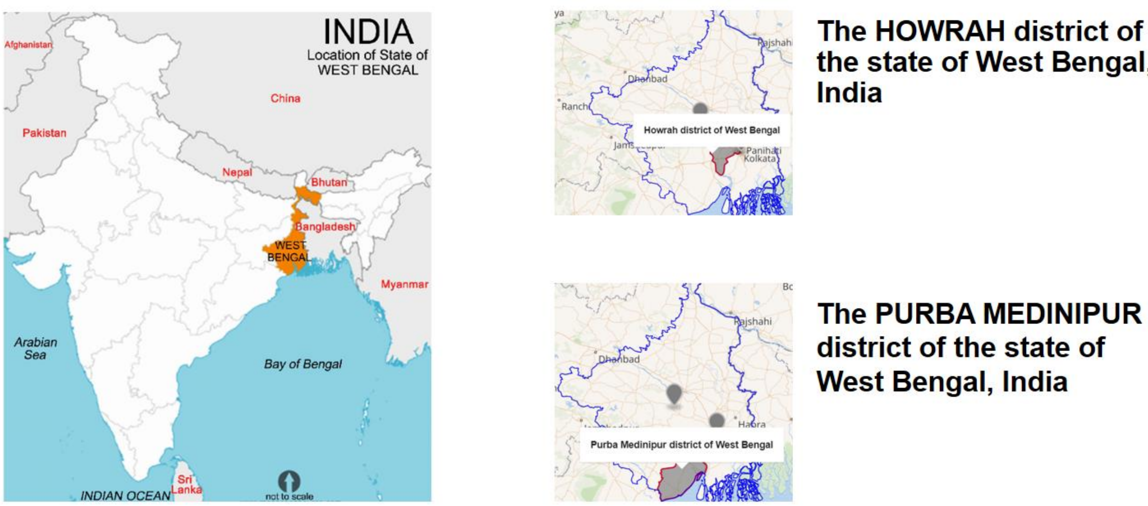


Figure 4 Location of Howrah and Purba Mehdinipur

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the project activity instances is the usage of traditional non-renewable wood fuels in inefficient cookstoves with poor ventilation in the kitchen, causing excessive indoor air pollution (IAP) and which has been a serious health risk for women and children who spend several hours in the kitchen for cooking as well as non-cooking activities.

And conventional wood fuel that is mostly non-renewable is used in traditional kilns to produce conventional charcoal; the latter being combusted in households and SMEs.

The proposed project activity reduces the use and demand for fossil fuels and non-renewable biomass that would have been used in the traditional cookstove to achieve the same output (i.e., cooking daily meals and boiling hot water) with the TLUD micro-gasifier. This directly leads in the reduction of GHG emissions from the cookstoves. Therefore, the project activity is implemented to generated GHG emissions reductions through clean cooking devices (TLUD micro-gasifier).

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no mandatory laws or regulations in the host country (India) requiring the introduction of gasifier in India, the project includes the distribution of Micro-gasifiers and is completely voluntary. The ongoing project does not violate any laws, statutes or other regulatory frameworks in India.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has not been registered and is not seeking registration under any other GHG emission program to avail carbon benefits during the crediting period of the project activity.

1.15.2 Projects Rejected by Other GHG Programs

The project proponent hereby corroborates that the project activity has not been rejected by any other GHG program

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The net GHG emission reductions generated by this grouped project activity will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions.

1.16.2 Other Forms of Environmental Credit

The project proponent hereby corroborates that the project activity has not created sought or received any other form of environmental credit.

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable for this grouped project, as the emission in supply chain is also taken in account for the calculation of the GHG emissions.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

Project's contribution to Sustainable Development-

The contributions of proposed grouped project activity towards sustainable development are explained with indicators viz. social, economic, environmental, technological well-being, legislative and temporal as follows:

- **Environmental well-being**

The project activity will result in the reduction of firewood consumption for cooking and charcoal production and emission of greenhouse gases and thus conserve forest and biodiversity.

- **Social well being**

The grouped project activity will pave the way for development and increase the social status living conditions and the prevailing living standard in the vicinity of the project activity thus resulting in empowering the nearby. Also, it will contribute to a small increase in local employment by employing skilled and un-skilled personnel for the operation and maintenance of the equipment.

The project will reduce the drudgery of women, timesaving and the use of saved time for other productive activities.

- **Economic well being**

The grouped project has created a business opportunity during the construction phase for local stakeholders such as suppliers, contractors etc. contributing to economic well-being aspects. Further, the project also influences the creation of employment opportunities for local people, which would enhance their social status. Sufficiently enhance indoor air quality thereby improving the health of women and children and reducing incidences of smoke and fire-related injuries and therefore result in saving of health-related expenses.

- **Technological well being**

The proposed project activity will promote TLUD micro gasifier hat [2]result in reduced fuel consumption and emissions due to cooking and heating water in homes.

- **Legislative:**

The Project Proponent has obtained all the relevant approvals required for the establishment and operation of the grouped project activity

[2] TLUD micro gasifier which is classified as "ACS" (advanced cook stoves) in VMR 0006 methodology

1.17.2 Sustainable Development Contributions Activity Monitoring

The project contributes to sustainable development in a number of ways:

a) Environmental Sustainability

   a. The project will help significantly reduce greenhouse gas emissions over its lifetime.

   b. The project will help reduce the use of non-renewable biomass from forests i.e., consumption of the wood fuel, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

b) Social Sustainability

   a. Considerably less time need to be spent collecting wood fuel for the family home thereby reducing the work burden on rural families and presenting alternative opportunities for economic development.

   b. The amounts of indoor pollutants from the burning of woof fuel in the family home will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total wood fuel burned and an increase in the temperature of combustion.

   c. The TLUD micro gasifier provide a safer method for combusting wood fuel for cooking, helping to reduce burn injuries, especially for children, in the family home

   d. Less time is required for the preparation of meal on TLUD micro-gasifiers as compared to the baseline traditional 3 stone mud cookstoves.

c) Economic Sustainability

   a. The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.

   b. Household expenditures on purchase of cooking fuel will be reduced through the use of the micro gasifiers.

   c. Saved household labor time which can be diverted to more productive economic activities. Hence, providing time for generating more livelihood options.

   d. Charcoal generated form the use of TLUD micro-gasifier is sold in the market hence generating an income for the beneficiary.

   e. The project will create local employment opportunities in operational and management roles, as well as future assembly and/or manufacturing initiatives.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	*SDG Target number*	*Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator*	*Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)*	*Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.*	*Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.*
1)	1.1	Household Income generated by the sales of the charcoal	Implemented activities to increase	Stove users can sell the charcoal generated in the stove during cooking to the project partner. Each month an additional revenue of Rs 271-300/- to the user is generated from the sales of charcoal.	This is a grouped project, it is planned to promote distribution and operation of 550,000 units of TLUD within 3 years. In accordance with pre-determined condition, the process of Charcoal Collection & Sales is carried out by project partner.
2)	3.9	User's perception of smoke reduction.	Implemented activities to decrease	Lowered the household air pollution of 2000 households by providing them TLUD micro-gasifier units	Lowered the indoor air pollution for the 550,000 households targeted for distribution of TLUD micro-gasifiers

3)	5.0	Achieve gender equality and empower all women and girls	Implemented activities to increase	By distribution of TLUD micro-gasifier units in the rural households, women and girls are empowered by technology upliftment, additional to it gasifier also generates sustainable charcoal which can be sold to the project implementer generating income, and the time required for cooking is also decreased by use of micro-gasifier.	Decrease in cooking time and wood collection time. And generating income for women and girls.
4)	7.1	Number of households that get access to affordable and reliable clean energy	Implemented activities to increase	In current monitoring period total of 2,000 TLUD micro-gasifiers ACS(Advanced Cook Stove) have been distributed to low-income households in rural area at an affordable, subsidized price through carbon financing."	Providing 550,000 of TLUD micro-gasifier units during the total project cycle of this grouped project.
5)	8.0	Creating job opportunities for semi-skilled and unskilled youth.	Implemented activities to increase	Assembly and distribution of TLUD micro gasifier is done by semiskilled workers. And Collection of charcoal from users, maintaining the stock of charcoal. "A total of 16 jobs were created."	is a grouped project, for 550,000 TLUD micro gasifier would be distributed. Into which 1100 people would be involved for distribution of cookstove while [number] would be involved in collection of the charcoal from the household level.

6)	12.0	12.2.2 Domestic material consumption, material consumption per capita, domestic material consumption	Implemented activities to decrease	Consumption of fuelwood for cooking purposes is decreased from 10kgs in baseline to 4.05 kg per day in a household.	Consumption of fuelwood in TLUD would be decreased as compared to baseline three stone/mud cookstoves
7)	13.0	13.2.2 Total Green-House gas emissions per year	Implemented activities to decrease	Emission reduction through the usage of TLUD micro gasifiers. For initial 2000 cookstove the decrease in ERs are 3,444 for duration of 4 months	Emission reduction through the usage of TLUD micro gasifiers is 21,393,963 tCO_2
8)	15.0	15.2.1 Progress towards sustainable forest management	Implemented activities to increase	Decrease in fuel wood consumption for the cooking purposes and charcoal making. In current monitoring period of 4 months the savings of fuelwood is 0.58 tonn per household and charcoal which was generated is 0.12 tonn per household	Decrease in fuel wood consumption for the cooking purposes and charcoal making. For each household 1.76 tonn of fuelwood is saved and generating 0.36 tonn of charcoal is generated in a year.

1.18 Additional Information Relevant to the Project

Leakage Management

There is no leakage for this project activity and project activity is already including 0.95 as a discount factor as mentioned on page number 9 of the methodology VMR0006 ver 1.1.

The group project does not involve any transfer of equipment from outside the project boundary to the project boundary. The project will use the micro-gasifiers which are manufactured within the project boundary.

Commercially Sensitive Information

Commercial sensitive information will be made available to VVB (Validation and Verification Board) on request.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

No negative impact of project activity. The project is implemented to reduce carbon emission from the inefficient burning of the firewood in three stone mud cookstove.

2.2 Local Stakeholder Consultation

The local stakeholder consultation for the grouped project activity was conducted on 27-November-2022 at Yogmaya restaurant, village- Bharitala, Jadurberia, Uluberia. The stakeholder consultation was conducted at the grouped project activity level, as the technology and its impact would be similar for all the instances which would be added later in the project.

Stakeholders from different social categories were invited, the stakeholders include

- Traditional cookstove user
- Women from villages
- women using the improved cookstove
- field assistants working in the project
- local NGO

Stakeholders were invited through public notices and invitation letters displayed. The notice for the meeting was also displayed at ATMs, Government buildings to get the maximum reach to the stakeholders.

A total of 33 participants attended the Local Stakeholder Consultation, and participants were encouraged to participate in the discussion.

A brief of stakeholder meeting is given below:

a) The project is to implement the 550,000 TLUD micro-gasifiers in India, and in initial phase 2,000 cookstoves are being distributed in West-Bengal.

b) The program is generating the carbon revenue which is in turn is used for subsidizing the cost of the gasifier for the user

c) The project will distribute the TLUD micro-gasifier.

d) The main aim of the project is to reduce the GHG emissions form the usage of traditional mud cookstove and protecting the trees of forest in the region. In addition, women get a livelihood option for the selling of charcoal produced at the end of cooking.

e) Process of the VERRA's validation and verification is explained, and process of regular audit was also explained to them.

f) Representatives were also introduced to the stakeholders, in case of any maintenance required in the cookstove, stakeholder can directly connect with the representative.

g) A trial for the cooking on the cookstove was also showcased among the stakeholder, and the operation of the cookstove is also demonstrated.

h) A point was suggested to prioritize the very poor household for the distribution of the gasifiers

Figure 5 Local Stakeholder Consultation

The Participants were provided with a feedback form to raise their concerns or comment on to the ongoing project. Overall, the project was easily understood and accepted by the different stakeholders and they were able to relate the benefits from using of TLUD micro-gasifiers and Improved cookstoves.

The table below list few of the comments and feedback received from the stakeholders:

Name of Stakeholder	Question Asked	Justification Provided
Lokhiramu Ghomu	When can we get this cookstove?	Initially 2000 TLUD are distributed, next lot will be distributed soon.
Shikha Gouri	How this cookstove work	The demonstration of cookstove and its principle of working was explained briefly
Mamoni Pradhan	How much wood is saved	On an average the TLUD required half of the wood required as compared to the traditional mud cookstove
Subroto Mandal	How can we participate in the project	Ground level field workers and staff are hired for distribution and monitoring of the cookstove.

For continuous input and monitoring of the project: the stakeholders were informed that the project has designed to continuous input and monitoring to ensure the usability of the cookstove by the beneficiaries. The contact details and location for maintaining the log book to record the query were also shared

Contact details for on-going communication

Sapient Infotech:

Mr. Moulindu Banerjee

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

After the detailed presentation on the grouped project the participants got involved asking them questions regarding the ICS.

- Environmental Benefits: reduction in the firewood consumption and emission of the Green House Gases, forest and biodiversity conservation
- Economic benefit: Employment creation and savings of the health cost.
- Air Quality: Sufficiently enhancing the Indoor Air Quality thereby improving health of women and children, and also reducing the incidences of smoke and fire related injuries in women

Social Benefits: reduces the drudgery of women, time saving and the use of the saved time for other livelihood activities.

2.4 Public Comments

There were no public comment received during the public comment period started from 16/01/2023 and ended on 15/02/2023.

2.5 AFOLU-Specific Safeguards

Project activity is not a AFOLU, hence this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Following approved baseline & monitoring methodology is applied.

Methodology:

VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves. Version: 1.1. Sectoral Scope: 03

AMS III BG- Methodology for Emission Reduction through Sustainable Charcoal Production and Consumption. Version 4.0. Sectoral Scope: 05

Tools:

CDM TOOL30: Calculation of the fraction of non-renewable biomass"; Version 04

CDM TOOL21: Demonstration of additionality of small-scale project activities";

3.2 Applicability of Methodology

S. No	VCS Methodology Requirement	Project Justification
Applicability for VMR0006 version 1.1		
01	Project activities shall be implemented in domestic premises, or in community-based kitchens.	The gasifier is distributed amongst the different household in the rural areas of India.
02	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	The thermal efficiency of the micro-gasifier comes out to be 28%- which is certified by third party. The usage of the micro-gasifier is reducing the consumption of the non-renewable biomass i.e., wood fuel used in the baseline.
03	Both 'Projects' and 'Large Projects' can use this methodology	The project is a large-scale project. It is categorised in 'Large Project' category. And this methodology is applicable for the current instance.
04	Non-renewable biomass has been used in the project region since 31 December 1989,	Proof that biomass has been used in the project region, i.e., in the

	using survey methods or referring to published literature, official reports or statistics	host country India since 31 December 1989 is available through widespread documentation. According to National Family Health Survey in year 2019-2020, in India around 42% of total household still uses non-clean fuel for their daily cooking practise. And in rural area this number goes up to 57%.[3]
05	For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	The gasifier is introduced as an energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass i.e., wood fuel for combustion. a) The consumption of the fuel used in the project activity will be monitored.
06	As the requirements of AMS II G version 11.1 paragraph 8: The project description shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).	A record-keeping system and unique identification of the project device will be used. This will include username, user identification number and address/ location of the user's house and distribution date. The record-keeping system will ensure that each TLUD micro gasifier can be traced to project instances to avoid double counting.
07	As the requirements of AMS II G version 11.1 paragraph 9: The project description shall also explain how the proposed procedures prevent double-counting of	The project proponent will sign an end-user agreement with individual households/users to ensure only

[3] India.pdf (rchiips.org) Indicator number 10

	emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	PP can claim the emission reduction. A default distribution agreement for end-users including the provision that emission reductions generated by the stove are owned by the PP will be provided for each user. A designated serial number is also used on each of the cookstove for its identification and prevent the cookstove device for the double counting in any other project.

Applicability for AMS III BG version 4.0		
S. No	**VCS Methodology Requirement**	**Project Justification**
1	Para. 2: This methodology is applicable to project activities that displace the use of non-renewable biomass in the production of charcoal supplied to identified consumers for thermal applications included in the project boundary.	Not applicable since project meet the criteria mentioned in paragraph 3 of the applicability conditions is applicable; and the charcoal is produced by micro gasifier stoves as a by-product of combustion.
2	Para. 3: This methodology is also applicable to charcoal generated as a by-product in micro-gasifier stoves using woody biomass for households cooking when used in conjunction with “AMS-II.G: Energy efficiency measures in thermal applications of non-renewable biomass”. Auxiliary power consumption in a blower or fan for forced convection is not covered by the methodology	The project is designed explicitly for the distribution of micro-gasifier stoves using firewood for cooking and not using auxiliary power in blowers or fans, such as the TLUD technology~~.~~ As mentioned in the methodology AMS III BG version 04.0. the emission on account of transport are assumed to be negligible
3	Para. 4: End users of charcoal shall be: (i) households; or (ii) small and medium enterprises (SMEs); or (iii) a group of households served by a charcoal market (e.g. charcoal consuming urban areas). End users do not include large scale industries.	End users for the charcoal are SMEs, will be identified during monitoring of parameter QCCP,i,y and evidence will be provided that they belong to categories (i), (ii) or (iii) as mentioned in Para. 4 of

	Footnote 1: Acceptable evidence include, but are not limited to: sales records and receipts of delivery of charcoal products directly to eligible end-users, long-term contracts with an entity (retailer, cooperative, trader etc.) supplying charcoal products to the eligible end-users.	AMS-III.BG, excluding large-scale industries. Evidence will include, but is not limited to: sales records and receipts of delivery of charcoal products directly to eligible end-users, long-term contracts with an entity (retailer, cooperative, trader etc.) supplying charcoal products to the eligible end-users. Evidence to show that end users are SME will be for example an undertaking of the charcoal traders that end users are SMEs.
4	Para. 5: Measures such as contractual agreements shall be implemented to avoid potential double counting because of potential claims of emission reductions by the end users. These measures shall be described in the Project Design Document	Charcoal users will sign agreements stating that they will not claim ER for the use of the charcoal provided under this project a designated serial number is also used on each of the cookstove for its identification and prevent the cookstove device for the double counting in any other project
5	Para. 6: Project activity, except for the case indicated in paragraph 3 above, shall introduce efficient charcoal production technologies using biomass feedstock such as biomass residues to displace the production of charcoal in unimproved traditional kilns by the informal sector thereby leading to emission reductions. Charcoal production facility may include briquetting facility for the agglomeration of smaller biomass particles. Methane produced during charcoaling process is either: (a) captured and destructed or gainfully used for heat or electricity; or (b) not captured and not destructed. Examples of these technologies include but are not limited to: (a) Retort sedentary kilns2 which	Not applicable since para. 3 is applicable; para. 6 refer to charcoal production in improved kilns, while the project is about charcoal generated in micro gasifier stoves.

	capture the pyrolysis gas; captured gas may be gainfully used for example as a fuel for pre-heating the facility or for wood drying or for production of heat and/or power; (b) Improved sedentary kilns without the capture of pyrolysis gas; (c) Casamance kilns.	
6	Para. 7 Project kilns not equipped with capture and destruction of the pyrolysis gases are not eligible to claim emission reductions on account of avoidance of methane emissions from the project activity under this methodology. It is assumed that methane emissions in the project equals to methane emissions in the baseline charcoal generation process.	Not applicable since para. 3 is applicable; para. 7 refer to charcoal production in improved kilns, while the project is about charcoal generated in micro gasifier stoves.
7	Para. 8: The project activity shall install and operate new (Greenfield) charcoal production facilities characterized by a new investment; replacement and retrofit of existing facilities is not eligible under this methodology. Provisions of “General guidelines for SSC CDM methodologies” shall be applied to demonstrate that the most plausible baseline scenario is the production of charcoal in unimproved traditional kilns by the informal sector.	Not applicable since project is about charcoal generated in micro gasifier stoves.
8	Para. 9: Charcoal manufacturing equipment transferred from existing or decommissioned charcoal production facilities are not eligible.	Not applicable since the project is about charcoal generated in micro gasifier stoves.
9	Para. 10: The biomass utilized by the project activity shall not be chemically processed, not stored for more than one year and not stored in anaerobic conditions.	Not applicable since no biomass is used explicitly for charcoal production; charcoal generated under this project is a by-product from cooking. The wood fuel is used in the micro-gasifiers, which are sourced from the agricultural land and nearby forest, which are kept in open and used within a week.

10	Para. 11: Biomass used by the project facilities is not stored for more than one year. No storage of the biomass is done in anaerobic conditions.	Not applicable since no biomass is used explicitly for charcoal production; charcoal generated under this project is a by-product of clean cooking. The biomass used in the project is been used from the agriculture land and other sources, the wood fuel which is harvested only last long for a week.
11	Para. 12: The embedded energy in charcoal produced as by-product in micro-gasifier stoves shall be neglected when performing water boiling test as per AMS-II.G. (see paragraph 17 of AMS-II.G., version 6) to ensure that efficiency estimates are conservative.	the charcoal produced in the micro-gasifier is not considered for the evaluating the stove efficiency.

S. No	CDM Tool 30 Requirement	Project Justification
Applicability for tool 30 version 04.0		
1.	DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or	The project location is India. This tool is used to calculate the f_{NRB} values for the same. The f_{NRB} value would be used for determining the project emission reductions.

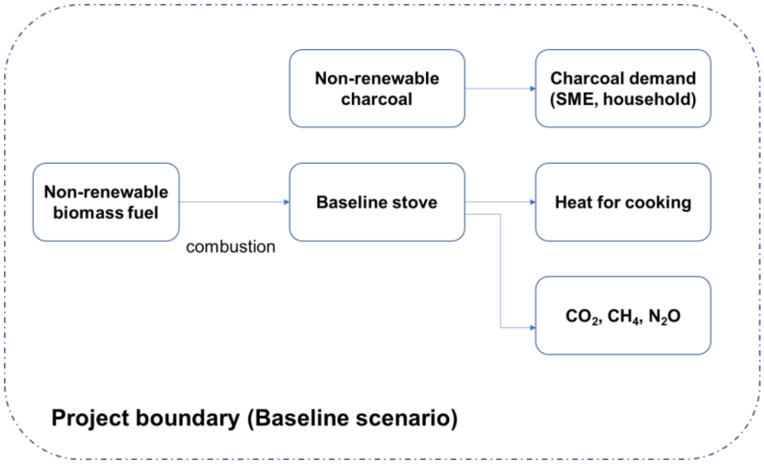
S. No	CDM TOOL 21 Requirement	Project Justification
Applicability for tool 21 version 13.1		
1.	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	Not applicable as no new methodology is not proposed.

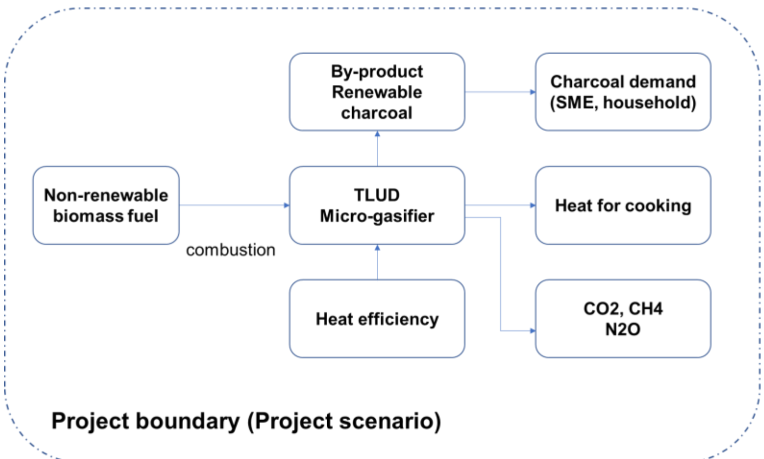
2.	Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	The tool is used for determination of the additionality of the micro-gasifier distribution project using the applicable methodology VMR0006 version 1.1 Not Applicable as the Tool-19 is only applicable for the micro-scale project.

3.3 Project Boundary

Grouped Project is implemented in the Republic of India, and sole motive is to provide a better cooking solution and reducing the Green House gas emission from the usage of the baseline cookstove.

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass cookstove	CO_2	Yes	Major Source
		CH_4	Yes	Major Source
		N_2O	Yes	Major Source
		Other	No	No other Source Identified
Project	Emission from the use of Micro-Gasifier for the cooking purposes	CO_2	Yes	Major Source
		CH_4	Yes	Major Source
		N_2O	Yes	Major Source
		Other	No	No other Source Identified





According to AMS III BG

Project boundary includes

(a) The use of biomass

(b) The carbonization units (including the micro-gasifier) included in the project;

(c) The areas of storage, processing, bagging and weighting of inputs (biomass) and outputs (charcoal)

(d) The use of charcoal and charcoal products.

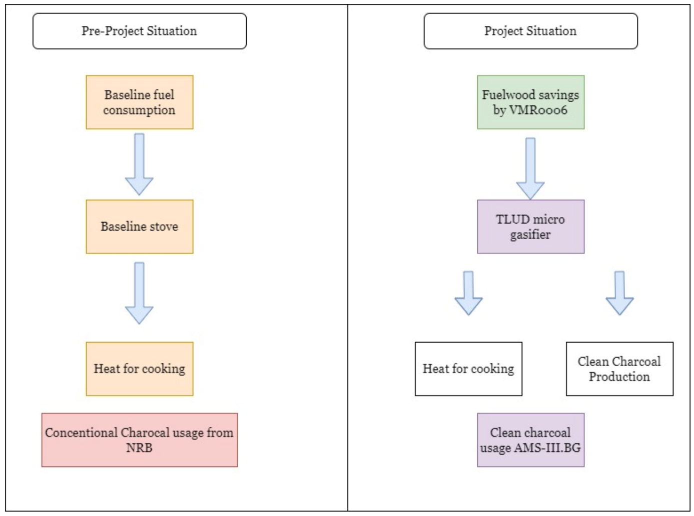


Figure 6 Representation of baseline boundary and Project Boundary

3.4 Baseline Scenario

The baseline scenario for cookstove component is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

A baseline survey is conducted in the project area from 18/08/2022 to 23/08/2022 and total of 379 households were interviewed by the team of 12 surveyors, for estimating the use of traditional cookstove in the areas, and it was found out that 95% of people in area still uses traditional cookstove for cooking and water heating purposes. [4]

As per the National Family Health Survey - 5 (2019-2020) for West Bengal[5], majority of households in rural areas are still using firewood for their daily thermal needs for cooking. In which more than 95% of household uses traditional cookstoves (chullah) for cooking meals.

Use of traditional mud cookstoves for cooking and boiling water by the project stove users is deemed as the realistic pre-project scenario. Traditional biomass i.e., wood fuel continues to be the primary energy source for cooking in rural areas and impoverished urban clusters. Approximately two-thirds of households rely on solid cooking fuels, including firewood, crop

[4] NFHS-5_INDIA_REPORT.pdf (rchiips.org)

[5] National Family health survey, the use of firewood for cooking can be determined from Table 4, page number 40. http://rchiips.org/nfhs/NFHS-5Reports/West_Bengal.pdf

residue, cow dung cake, and coal/lignite/charcoal[6]. According to the baseline survey conducted by the Project Participant, inefficient cooking practices are widespread in rural areas of West Bengal.

The survey revealed that households using fuelwood as a fuel for the cooking purposes. It's crucial to note that household air pollution is closely linked to a heightened risk of various health issues, including tuberculosis, asthma, mortality from cardio-respiratory illnesses, and non-respiratory problems such as adverse pregnancy outcomes, prematurity, and low birth weight[7]. The survey also shows that most of the fuelwood is purchased from the market and around one-fourth of the wood is being collected from the nearby sources.

At the national level, statistics provided by the National Institution for Transforming India (NITI Aayog) offer an overview of the baseline energy scenario in India. These statistics underscore the pressing need for interventions and improvements in cooking technologies and energy sources, especially in rural areas, to mitigate the adverse health effects associated with traditional biomass use.

The proportion of population using solid fuels in India was 55·5% (54·8–56·2) in 2017 and 1·24 million (1·09–1·39) deaths in India in 2017, which were 12·5% of the total deaths, were attributable to air pollution[8]. The Project activity are premised on generating Emission Reductions by ensuring that users have cleaner cookstoves, thereby reducing the need for them to burn non-renewable biomass in order to cook food.

There is currently no law or policy which requires the use of fuel-efficient stoves in India. It follows that the Project activity is a voluntary action.

As per the baseline survey conducted in the region, the key reason for current cooking practice of traditional 3-stone chulhas are: easy availability of wood from forest, lack of awareness about the fuel wastage and other harmful effects of low efficiency stoves and the capital cost of the efficient technologies.

The baseline scenario for charcoal generation component (AMS-III.BG), the project aims at replacing conventional charcoal used by different consumers eligible under AMS-III.BG such as SMEs consuming charcoal (barbecue restaurants). Such users traditionally use conventional charcoal produced from unsustainably harvested fuelwood and in project case the SMEs are consuming the charcoal produced by the use of micro-gasifiers. The charcoal in baseline was

[6] Kumar, P., & Igdalsky, L. (2019). Sustained uptake of clean cooking practices in poor communities: Role of social networks. Energy research & social science, 48, 189-193.

[7] Jindal, S. K., Aggarwal, A. N., & Jindal, A. (2020). Household air pollution in India and respiratory diseases: current status and future directions. Current Opinion in Pulmonary Medicine, 26(2), 128-134

[8] Balakrishnan, K., Dey, S., Gupta, T., Dhaliwal, R. S., Brauer, M., Cohen, A. J., & Sabde, Y. (2019). The impact of air pollution on deaths, disease burden, and life expectancy across the states of India: the Global Burden of Disease Study 2017. The Lancet Planetary Health, 3(1), e26-e39

produced from non-renewable biomass in the baseline and in absence of the of the project activity the baseline scenario would be the future use of fossil fuels for meeting the similar thermal energy needs.

The baseline for renewable biomass based charcoal generation is demonstrated as follows;

a. The biomass being renewable: The fraction of Non-Renewable Biomass (NRB) is calculated and (1-NRB) is the Renewable biomass. There is no change in the baseline production process i.e., open ended method for the charcoal manufacturing which is unsustainable and it is supported by the report published by Indian Institute of Forest Management, on The Feasibility of Wood Based Charcoal in Industrial Processes[9], where in section 4.3 of report concludes the production process of the charcoal being unsustainable. This is also concluded in the report on the Charcoal as Global commodity: it is sustainable[10], by UN Environment Program.

   And for the emission reductions taken into account in equation (1) of the methodology AMS III BG version 04.0 uses the default values, thus calculations are conservatively estimated.

3.5 Additionality

For the VMR0006 version 1.1 methodology, project uses the activity method and project method for the demonstration of additionality

1: Regulatory Surplus

There is no mandated government programme or policy in the host country (India) of this project ensuring the distribution of domestic fuel-efficient gasifier. The project is not mandated by any law, statute, or another regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory frameworks. Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by Sapient Infotech in different regions of India.

Step 2: Positive List and simple cost analysis.

The micro-gasifier will be distributed to the end-user at a subsidized rate of 1250, hence there will be a revenue stream apart from the sale of GHG credits. Therefore, PP chose option 3 for demonstration of additionality. The project is not implemented as part of government schemes or supported by multilateral funds. The project did not apply any methodology deviations.

[9] Report by IIFM on The Feasibility of Wood Based Charcoal in Industrial Process, Section 4.3, page number 44. https://moef.gov.in/wp-content/uploads_3/2018/08/Charcoal-Final-Report-20-March-for-Print-2.pdf

[10] Report by UN Environment Program, Report title: Charcoal as a global commodity: is it sustainable? https://wedocs.unep.org/bitstream/handle/20.500.11822/40469/CHARCOAL.pdf

1- Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology.

Step 3: Project Method

Additionality for Instance-1

In line with Methodology VMR0006: Version 1.1, para 7 step 3 "For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible"

Hence, in line with AMS-II.G. Version 13.0 section 5.2; additionality is demonstrated using Option 2, by applying the "TOOL21: Demonstration of additionality of small-scale project activities"; section 5, i.e., *Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions*.

PP has identified realistic and credible alternative scenario(s) to the project activity. The alternative identified are

(a) The proposed project activity (i.e., distribution of micro-gasifier that are efficient and have lower emissions) undertaken without being registered as a VCS project activity;

(b) Continuation of the current situation (i.e., continuation use of traditional stoves which are free of cost but have higher emissions).

Investment Barrier:

Alternative (a) i.e., proposed project activity:

The proposed project intends to distribute the ICS at 58% subsidy to users i.e., households; hence financial return from the programme will be less than cost incurred.

A sum of 1,250 Indian rupees, is collected from each of the beneficiary for the purpose of a partial contribution towards the cost of micro-gasifier and to ensure effective implementation of project. There is no other revenue generated in the project by the project proponent other than the sale of GHG credits. Moreover, the distribution cost is documented in the agreement between PP and Distribution Agency.

The recovered cost is clearly documented in the user agreement between PP and micro-gasifier beneficiaries. This can be verified in the user agreement attached in the *Appendix II: Sample of end user Agreement* of this document.

Further each micro gasifier has certain costs associated with the production/purchase, distribution, repairs and maintenance, project management, monitoring and verification etc., the programme will not occur in absence of the carbon revenue.

Hence a **simple cost analysis** is chosen by PP.

PP has carried out simple cost analysis for the demonstration of additionality and from above explanation it is very clear that the ICS has been distributed at a 68.75% subsidised cost to the end-user or beneficiaries. Hence the proposed project activity is not economically or financially feasible without the revenue from sale of GHG credits.

The below table shows the revenue and subsidy for the Improved cookstove

S. No	Particular	Cost (INR)	Source
01	Hardware cost	3,100	Invoice
02	Cost of distribution	900	Contracts
03	Cost paid by beneficiary	1,250	Contract with distribution agency.
04	Percentage subsidy to beneficiary	68.75%	Calculated

The action is not financially viable without the support of revenues from the sale of VCUs. As the end-user does not benefit from a direct financial return and procuring micro-gasifier requires capital, which is a barrier to rural consumers due to difficulties in accessing capital, a wide dissemination of micro-gasifier in the Host Country is unlikely. The actions under the project will alleviate these barriers by promoting distribution of micro-gasifiers to end-users at an affordable price.

Alternative (b) i.e., Continuation of the current situation:

Continuation use of traditional stoves which are free of cost and financially more viable alternative which leads to higher emissions.

Hence, the project faces investment barrier and is considered to be additional.

Conclusion:

From the above explanation, it is very clear that the end users have a financially more viable alternative to the project activity i.e., continuation use of traditional stoves which are free of cost but have higher emissions. Hence, it can be concluded that the project faces Investment Barrier.

For the demonstration of additionality of charcoal component adhere to the AMS III BG version 4.0. methodology, project is applying the "TOOL21: Demonstration of additionality of small-scale project activities"; section 5, i.e. Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions.

PP has identified realistic and credible alternative scenario(s) to the project activity. The alternative identified are

(a) The proposed project activity (i.e., distribution of micro-gasifier and implementation of charcoal collection, storage and sales) undertaken without being registered as a VCS project activity;

(b) Continuation of the current situation (i.e., the charcoal production from informal sector with use of non-renewable biomass in the production of charcoal supplied to identified consumers for thermal applications included in the project boundary.)

Investment Barrier:

Alternative (a) i.e., proposed project activity:

As set out in para 3.6.2 of VCS standard version 4.4, separate demonstration of additionality may not be practicable in project activities that are implemented at a single facility and therefore represent a single investment.

TLUD gasifier simultaneously function as an advanced cookstove and sustainable charcoal production facility. As Charcoal production under this project is not available without distribution of TLUD.

As calculated above, net investment of per TLUD shall be INR 2,750/TLUD. The current charcoal value chain of the project is operated based on fixed price basis, with payment to TLUD end users and selling sustainable charcoal to contracted retailer. The project proponent does not have share in the revenue which is generated from the sales of charcoal. The part of the revenue is paid back to the beneficiary as sale of charcoal produced by the use of micro-gasifier, and remained share of the revenue is used in maintaining the supply chain of charcoal collection from different beneficiary and transporting it to the vendor. Condition is determined with the consideration on contribution to SDGs and based on the premise that the aforementioned investment cost is covered by carbon financing and the fixed terms reflects

Commercially confidential information will be adequately verified through site visit and review on contracts or other documents by VVB, and economic viability for investment on TLUD distribution as the charcoal production facility cannot be secured without carbon financing.

Alternative (b) i.e. Continuation of the current situation:

As per para 18. Definitions of AMS III BG version 4.0., informal charcoal sector is referred to as individuals or a group of individuals involved in charcoal production but are not formally registered or formally charged with production and supply of charcoal products or related service by the authorities. It is characterized by the use of traditional kilns such as earth mound kilns, pit kilns or equivalent open-end technologies which require no investment besides labour. Newly established formalized organization by such individuals, e.g. cooperative, can also be considered as the informal sector for the purpose of this methodology;

Continuation use of charcoal production, which are free of cost and financially more viable alternative which leads to higher emissions.

Hence, the project faces investment barrier and is considered to be additional.

Conclusion:

From the above explanation, it is very clear that the producer for charcoal supplied to the end-users included in the project boundary have a financially more viable alternative to the project activity i.e., continuation of charcoal production from informal sector, which are free of cost but have higher emissions. Hence, it can be concluded that the project faces Investment Barrier.

3.6 Methodology Deviations

The project activity has not taken any methodology deviation in the applied approved methodology VMR0006 version 1.1, AMS III BG version 4.0.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The distribution of the micro-gasifier under this grouped project has been started. The distribution would be done in different instances, and in each instance the different batches of the micro-gasifier or cookstoves would be distributed.

In a batch the model of the micro-gasifier or improved cookstove would remain same and the details about the micro-gasifier and improved cookstove would be mentioned in the table below

Instance	Phase	Model of micro-gasifier	Number	Distribution date
Instance 1	Batch 1	TLUD micro gasifier	2000	From 10 September 2022 to 25-September-2022

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The methodology VMR0006 follows AMS II G and which does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. Please refer to Section 5.4.

For the methodology AMS III BG, paragraph 20 states the baseline emissions of the project, it is assumed that in the absence of project activity, the baseline scenario would be the future use of fossil fuels for meeting up the similar thermal energy needs.

And in paragraph 21, it is said that for charcoal production from renewable biomass in baseline, traditional open-ended methods resulting in the methane emissions to the atmosphere from the baseline scenario.

5.2 Project Emissions

The methodology VMR0006, does not account for project emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. Please refer to Section 5.4.

The methodology AMS III BG, the project emission during the use of micro-gasifier cookstove is taken is considered. There are five different types of Project emissions are mentioned as:

$PE_{y,fugitive}$ = Fugitive emissions from operation of charcoal producing facility (physical leakage) in the year y (t CO2e)

This has been calculated separately as per the equation (2) of the applied methodology AMS III BG version 04.0

$$PE_{y,fugitive} = \sum_i Q_{CCP,i,y} \times GWP_{CH4,y} \times SMG_{y,b} \times f$$

This equation is combined with the Emission Reductions equation (1).

$PE_{y,flaring}$ = If applicable, emissions due to the flare inefficiency in the project charcoal manufacturing plant in the year y (t CO2e) determined in accordance with the procedure provided in AMS-III.K. In case captured pyrolysis gas is gainfully used (e.g. as fuel for pre-heating the facility, or for wood drying, or used for production of heat and/or power as in the case of micro-gasifier), then it can be taken as zero

In the project activity; there is use of micro-gasifier for the production of the

Charcoal. Therefore, the flaring emission is taken as zero.

$PE_{FF,y}$ = Project emissions due to fossil fuel consumption in charcoal production facilities in year y (t CO_2)

There is no use of the fossil fuel for the charcoal production, because charcoal is been produced as a by-product of the combustion of the firewood inside the TLUD stove. The heat of the pyrolysis is used for the cooking purposes and at the end, the remaining is a by-product in form of charcoal. Therefore, the project emissions due to fossil fuel consumption in the charcoal production facility is taken as zero.

$PE_{El,y}$ = Project emissions due to electricity consumption in charcoal production facilities in year y (t CO_2)

There is no use of the electricity consumption for the production of charcoal. In the project activity charcoal is produced as a by-product of the combustion of the firewood inside the TLUD stove. The heat of the pyrolysis is used for the cooking purposes and at the end, the remaining is a by-product in form of charcoal. Therefore, the project emissions due to electricity consumption in the charcoal production facility is taken as zero

$PE_{BC,y}$ = Project emissions due to biomass cultivation in year y (t CO2)

The biomass used in the micro-gasifier is the wood fuel, which is collected or purchased from the market as similar in the baseline scenario. There are no other emissions which are involved for the biomass cultivation. Therefore, the project emissions due to biomass cultivation is taken as zero.

5.3 Leakage

As per the section 8.3 of the applied methodology; VMR0006 version 1.1; Leakage has considered as a default value of 0.95 in accordance with Section 5.4 of AMS II G

According to section 5.4 of applied methodology AMS II BG version 04.0; General guidance on the leakage in biomass project activities shall be followed to quantify leakages pertaining to use of biomass residues. For this as per the section 5.4 of AMS II G; Leakage has considered as a default value of 0.95.

5.4 Estimated Net GHG Emission Reductions and Removals

The usage of micro-gasifier has incorporated two of the methodologies VMR0006 and AMS III BG. the different equation from the methodology is used to determine the overall GHG emission reduction from the project

The micro-gasifier is introduced as energy efficiency measure in the project, therefore equation 1 and 2 of the methodology is employed for the calculation of GHG emission reductions

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Equation (1)

Where:

i = indices for the situation where more than one type of cookstove is introduce to replace three stone fire

j = indices for the situation where there is more than one batch of the improved cookstove of type i

ER_y = emission reduction during the year y in tCO_2e

$ER_{y,I,j}$ = Emission reductions by the improved cookstoves of type i and batch j during the year y in tCO_2e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times \left(EF_{wf,CO2} + EF_{wf,non\ CO2}\right) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where,

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y

$f_{NRB,y}$ = Fraction of woody biomass that can be established as a non-renewable biomass (f_{NRB})

$NCV_{wood\ fuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default value for wood fuel, 0.0156 TJ/tonne)

$EF_{wf,\ CO2}$ = CO_2 emission factor for the wood fuel in baseline scenario (IPCC default value for wood fuel, 112 tCO_2/TJ)

$EF_{wf,\ non\text{-}CO2}$ = Non-CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default value for wood fuel, 26.23 tCO_2/TJ)

$N_{y,i,j}$ = Number of improved cook stoves of type i in batch j operating during the year y

0.95 = discount factor to account for leakage

The quantify of woody biomass saved $B_{y,saving,i,j}$ due to implementation of improved cook stoves can be estimated by Equations 4:

$$\boldsymbol{B_{y,savings,i,j}} = \boldsymbol{B_{y=1,new,i,survey}} \times \left(\frac{\boldsymbol{\eta_{new,i,j}}}{\boldsymbol{\eta_{old}}} - \mathbf{1}\right)$$ Equation (4)

Where,

η_{old}	=	Efficiency of baseline stove
$\eta_{new,i,j}$	=	Efficiency of the improved cookstove of type i and batch j determined through the water boiling test (WBT). Alternatively, efficiency may be determined using the equation (5) of the applied methodology PP is choosing the method set out in the equation 5 of the applied methodology.
$B_{y=1,new,i,j,survey}$	=	Annual quantity of woody biomass used by improved cook stoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Equation (5)

η_p	=	Efficiency of project stove (fraction) at the start of project activity
$(DF_n)^{y-1}$	=	Discount factor to account for efficiency loss of project cook stove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards Alternatively default value of 0.99 efficiency loss per year can be considered PP is choosing the factor of 0.99 efficiency loss per year.
0.94	=	Adjustment factor to account for uncertainty related to project cook stove efficiency test

AMS III BG

For the project technology equipped with capture and destruction of the pyrolysis gases, including micro-gasifier, emission reductions are calculated as follows:

$$ER_y = \sum_i Q_{CCP,i_y} \times \left[\left(CF \times NCV_{wood} \times \frac{NCV_{charcoal,i}}{NCV_{charcoal,default}} \times f_{NRB,BL,wood} \times EF_{projected\_fossilfuel}\right) + (SMG_{y,b} - M_d) \times (1 - f_{NRB}) \times GWP_{CH4,y}\right] - PE_{y,fugitive} - PE_{y,flaring} - PE_{FF,y} - PE_{El,y} - PE_{BC,y}$$

Equation (1)

Where:

ER_y = Emission reductions in year y (t CO_2e/yr)

$Q_{ccp,iy}$ = Quantity of charcoal type i produced and used in year y (t)

CF = Wood to charcoal conversion factor

NCV_{wood} = Net calorific value of wood (TJ/t)

$NCV_{Charcoal,i}$ = Net calorific value of the charcoal type i produced during the project (TJ/t)

$NCV_{Charcoal,default}$ = Default net calorific value of charcoal (TJ/t)

f_{NRB} = Fraction of biomass used in the absence of the project activity that can be established as non-renewable biomass; (fraction or %)

$EF_{project\_fossilfuel}$ = Emission factor for the substitution of non-renewable woody biomass by similar consumers (t CO_2/TJ)

$GWP_{CH4,y}$ = Global warming potential of methane applicable to the crediting period (t CO_2e/t CH4)

$SMG_{y,b}$ = Specific methane generation for the baseline charcoal generation process in the year y (tonnes CH4/t charcoal product); a default value of 0.030 t CH4/t charcoal may be used. Alternatively, the value can be determined in accordance with the procedure provided in the latest version of “AMS-III.K: Avoidance of methane release from charcoal production”

M_d = Factor to account for any legal requirement for capture and destruction of methane in the charcoal production facility (tonne of CH4/tonne of raw material)

$PE_{y,flaring}$ = If applicable, emissions due to the flare inefficiency in the project charcoal manufacturing plant in the year y (t CO_2e) determined in accordance with the procedure provided in AMS-III.K. In case captured pyrolysis gas is gainfully used (e.g. as fuel for pre-heating the facility, or for wood drying, or used for production of heat and/or power as in the case of micro-gasifier), then it can be taken as zero

$PE_{FF,y}$ = Project emissions due to fossil fuel consumption in charcoal production facilities in year y (t CO_2)

$PE_{EL,y}$ = Project emissions due to electricity consumption in charcoal production facilities in year y (t CO_2)

$PE_{BC,y}$ = Project emissions due to biomass cultivation in year y (t CO_2)

$PE_{y,flaring}$ is not applicable since pyrolysis gas is used for cooking.

M_d is set to zero since there is no legal requirement to capture methane in micro gasifier stoves.

$PE_{FF,y}$ and $PE_{El,y}$ are not considered since no fossil fuels or electricity are used in the micro gasifier stoves;

$PE_{BC,y}$ is not applicable since no biomass will be cultivated for charcoal production since it is produced as a by-product of daily cooking.

$PE_{y,fugitive}$ is calculated as follows :

$$PE_{y,fugitive} = \sum_i Q_{CCP,i,y} \times GWP_{CH4,y} \times SMG_{y,b} \times f$$

Equation (2)

Where:

$PE_{y,fugitive}$ = Fugitive emissions from operation of charcoal producing facility(physical leakage) in the year y (t CO_2e)

f = A fraction attributed to project charcoal production technology, use a default value of 0.1.

Equations 1 and 2 can thus be combined and simplified

$$ER_y = \sum_i Q_{CCP,iy} \times \left[\left(CF \times NCV_{wood} \times \frac{NCV_{charcoal}}{NCV_{charcoal,default}} \times f_{NRB,BL,wood} \times EF_{projected\_fossilfuel}\right) + SMG_{y,b} \times (0.9 - f_{NRB}) \times GWP_{CH4,y}\right]$$

Equation (3)

$f_{NRB,y}$ is determined by using tool 30 as described in monitoring section under VMR0006

The fraction of woody biomass that can be established as non-renewable is

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where,

NRB = Quantity of non-renewable biomass consumed in the applicable areas in relevant period (tonnes)

RB = quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

$$NRB = H - RB$$

Where

H = Total consumption of woody biomass in the applicable area in the relevant period. (tonnes)

$$H = HW \times N + CE + NE$$

Where;

HW	=	Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes//household)
CE	=	Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)
NE	=	Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)
N	=	Number of households consuming wood fuel within the applicable area in the relevant period (number)

$$RB = \sum \left(MAI_{forest,i} \times \left(F_{forest,i} - P_{forest,i} \right) \right) + \sum \left(MAI_{other,i} \times \left(F_{other,i} - P_{other,i} \right) \right)$$

$MAI_{forest,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period (tonnes/ha/yr)
$MAI_{other,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category i of other land areas in the relevant period (tonnes/ha/yr)
$F_{forest,i}$	=	Extent of forest in sub-category i in the relevant period (ha)
$F_{other,i}$	=	Extent of other land in sub-category i in the relevant period (ha)
$P_{forest,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category i) in the relevant period (ha)

$P_{other,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category i) in the relevant period (ha

i = Sub-category i of forest areas and other land areas

$$RB = 0.949(822{,}789 - 377{,}200) + 0.34(1{,}206{,}900 - 0)$$

$$\mathbf{RB = 837,710\ tonnes}$$

$$H = 5{,}394{,}057 + 0 + 4{,}282$$

$$\mathbf{H = 5,398,339\ tonnes}$$

H	Calculated	5,398,339
RB	Calculated	837,710
NRB	Calculated	4,560,629
FNRB		**84.48%**

Comparison of the f_{NRB} values calculated that too reported in the literature:

The significant difference between the values of f_{NRB} achieved by the Bailis report "The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272." (25.8%) and the values achieved by our project (84.48%) is indeed notable. To address the feasibility of the value calculated based on CDM Tool 30 version 4 and its alignment with the conservative principle in the VCS Standard section 2.2, the following points should be considered:

1. **Data Source and Accuracy**:

   VCS 3954: The project's f_{NRB} calculation relies on the most recent data of forest cover, fuelwood consumption, annual increment from the "Forest Survey of India" report published by Ministry of Environment Forest and Climate Change (MoEFCC). The report is published every two years and the latest report referred in the project is of 2021(FSI)[11]. While the demographic details (family size, household data using fuel wood for cooking) are based on the National Family Health Survey (NHFS) report, NFHS surveys have been conducted under the stewardship of the Ministry of Health and Family Welfare (MoHFW), Government of India. MoHFW has designated the International Institute for Population Sciences (IIPS) which publishes the survey report. The project applies the data from the latest available report of NFHS-5[12] 2021. The f_{NRB} value is calculated based on the above-mentioned reports specifically for the country India, i.e., project boundary.

[11] The data of the Forest Survey of India is published regionally as well as at the national level. The latest reports for India and the project boundary could be downloaded from https://fsi.nic.in/fsiwebsite/forest-report-2021

[12] National Family Health Survey is conducted every two years and the latest published data are provided on the link http://rchiips.org/nfhs/NFHS-5_State_Report.shtml . This report was published in the year 2021.

Bailis report: The report uses global-level data from the Food and Agriculture Organization (FAO), the International Energy Agency (IEA) and the United Nations (UN)[13] for various data along with other published literature for specific countries.

For specific f_{NRB} data for India, Bailis report estimates the NRB values of the region using the WISDOM Model[14] (as referred literature of the paper *"Drigo, R. & Salbitano, F. WISDOM for Cities: Analysis of Wood Energy and Urbanization Using WISDOM Methodology (FAO Forestry Department Report, 2008", Drigo, R. WISDOM Case Studies (2014); http://www.wisdomprojects.net/global/cs.asp", "Masera, O., Ghilardi, A., Drigo, R. & Trossero, M. A. WISDOM: A GIS-based supply demand mapping tool for woodfuel management. Biomass Bioenergy 30, 618–637 (2006)."*). which takes the values of supply and demand module of a given spatial base of the region, and reliance on the local surveys for the same was one of the parameters in the model. Since, there was no survey conducted in the region for the demand of the woodfuel, the Bailis report has referred to the published data of woodfuel consumption for India as per two reference literature;

a. Paper titled "*Comparative study of fuelwood consumption by villagers and seasonal ''Dhaba owners'' in the tourist affected regions of Garhwal Himalaya, India"* [15]. This report refers to wood fuel consumption for "Dhaba owners" i.e. small restaurants in local language. The region of study was tourist affected region of Garhwal which is in Uttarakhand state of India. This paper was published in year 2010.

b. Paper titled "*Firewood consumption along an altitudinal gradient in mountain villages of India*"[16]. This report refers to fuel wood consumption of village households in Uttarakhand state of India. This paper was published in year 2004.

Inference: The data used by Bailis report specifically for India is based on a paper of 2004 and 2010, thus not representing the latest data as published by the national agencies as well as regional data available for the project boundary, i.e., India. There is considerable difference in the values as published by the national agencies. Further Bailis report does not take into account the values for the project boundary, i.e. India.

2. **Adherence to Tool 30:** The project applies VMR0006 version 1.1 which refers the calculation of f_{NRB} based on latest applicable version of Tool 30[17] as published by CDM EB, i.e version 4.0. The tool provides detailed steps to calculate the f_{NRB} value for a given region.

   VCS 3954: The project has detailed step wise calculation for assessment of f_{NRB} values. The value has been calculated for the applicable project boundary i.e., West Bengal state in country India by using latest regional level data as published by the various government agencies. The f_{NRB} value arrived is 84.48% for the project boundary.

[13] As mentioned in the section 'Methods' of the paper: The carbon footprint of Traditional Woodfuels.

[14] Woodfuel Integrated Supply/Demand Overview Mapping (WISDOM) Model which estimates the value of the fuelwood consumptions in the areas. https://www.sciencedirect.com/science/article/abs/pii/S0961953406000262

[15] This study is based on the fuelwood consumption for the small restaurants of the Garhwal Region in Uttarakhand, this demand cannot be used for estimating the fuelwood consumption for the country. https://www.sciencedirect.com/science/article/abs/pii/S0301421509009197

[16]The woodfuel consumption for Uttarakhand state and that too variation in the consumption pattern as per altitude change. https://www.sciencedirect.com/science/article/abs/pii/S0961953403001909

[17] The https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

Bailis Report: The report is based on WISDOM Model. The f_{NRB} values published for India is 23-25%. The tool also published a default value of f_{NRB} value that can be used which is 30%.

Inference: the f_{NRB} value as calculated by the VCS 3954 is in adherence to the Tool 30 version 4.0.

3. Limitations of Bailis Report: As published in the Bailis report we would like to highlight the below:

   Section "Discussion and Implementation" on page number 4 mentions ***"One limitation of the study is a lack of reliable woodfuel consumption data. When possible, we used national and sub national data sets. However, for most countries, we relied on data compiled by international organizations containing unknown uncertainties that make it difficult to communicate the uncertainty in these results. A second limitation is that the analysis considers a single year and does not account for potential behavioural changes among wood fuel users"***.

   Inference: The Bailis report is not a correct representative of the f_{NRB} for the given instance boundary, i.e. state West Bengal in country India.

4. Values published in other projects: The f_{NRB} values of various other registered projects having similar project boundary, a summary of the same is mentioned below:

   VCS: details about other VCS projects:

VCS ID	f_{NRB} values	Date of Registration
2607	93%	19/01/2023
2533	90.10%	02/07/2022
2473	93.10%	09/06/2022

Gold Standard:

GS ID	f_{NRB} values	Date of Registration
PoA 916 and all VPAs	93.10%	
11358	80.13%	07/04/2022
PoA: 11450 and all its CPAs	87%	18/04/2021

Inference: The f_{NRB} values are comparable to other similar projects in the region, which has been registered by the Verra and Gold Standard Mechanism and verified by VVB. Therefore, the conservative principle does not apply in this context.

Conclusion: There are considerable differences in the approach and value determined by the project as compared to the Bailis report. Since the value of f_{NRB} as calculated in the project is based on latest specific regional area of country thus it represents actual scenario as compared to f_{NRB} as reported by Bailis report which is for the country. The report itself claims data to be from 2004 and 2010 thus it does not represent the actual scenario. The report fails to mention the uncertainty in the results. Further, the value is calculated for a single year and does not take into account the trend in the past decade.

We would thus like to mention that the f_{NRB} value as calculated in the project is appropriate for the given project boundary.

Conservative assessment: The comparison with Bailis report is not applicable since the data and uncertainty for f_{NRB} values in Bailis report are not reliable for the project boundary, i.e. India.

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
10 September 2022 – 31 December 2022	3,951	0	0	3,951
01 January 2023 – 31 December 2023	1,643,025	0	0	1,643,025
01 January 2024 – 31 December 2024	3,574,374	0	0	3,574,374
01 January 2025 – 31 December 2025	3,530,452	0	0	3,530,452
01 January 2026 – 31 December 2026	3,486,970	0	0	3,486,970
01 January 2027 – 31 December 2027	3,443,922	0	0	3,443,922
01 January 2028 – 31 December 2028	3,401,304	0	0	3,401,304
01 January 2029 – 09 September 2029	2,309,966	0	0	2,309,966
Total	**21,393,963**	**0**	**0**	**21,393,963**

6 MONITORING

6.1 Data and Parameters Available at Validation

For VMR0006

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity
Equations	4
Source of data	Manufacturer's specification
Value applied	28%
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. The project device is a champion TLUD micro gasifier. The specification of the same are provided in the Appendix II of the document.
Purpose of Data	Calculation of η_{new} and calculation of emission reductions
Comments	This parameter shall remain fixed for the crediting periods.

Data / Parameter	f_{NRB}
Data unit	Percentage (%)
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Equations	4
Source of data	Determined by using Tool 30: Calculation of the fraction of non-renewable biomass. Version:04.0
Value applied	84.48%
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of emission reductions
Comments	Parameter f_{NRB} once fixed shall remain fixed for the entire crediting period

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	The net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	Default as per applied approved methodology VMR0006 ver 1.1 Page number 11
Value applied:	0.0156
Justification of choice of data or description of measurement methods and procedures applied	The Default value as per methodology VMR0006, ver1.1, page number 11
Purpose of data	Calculation of emission reductions
Comments	This parameter is fixed ex-ante

Data / Parameter	η_{old}
Data unit	Fraction
Description	The efficiency of baseline cookstove
Source of data	default value
Value applied:	0.1
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.1 shall be used if baseline device is a three-stone fire using fuelwood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. This is verified during the baseline survey conducted before the implementation of the project.
Purpose of data	*Calculation of emission reductions*
Comments	This value is fixed ex-ante

Data / Parameter	$EF_{wf,\ CO2}$
Data unit	tCO_2/TJ
Description	CO_2 emission factor for the use of wood fuel in the baseline scenario
Source of data	Default value as per methodology VMR0006, v1.1, Page no. 10
Value applied:	112

Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

Data / Parameter	$EF_{wf, non\text{-}CO2}$
Data unit	tCO_2/TJ
Description	Non-CO_2 emission factor for the use of wood fuel in baseline scenario
Source of data	Default value as per methodology VMR0006, v1.1, Page no. 10
Value applied:	26.23
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per methodology VMR0006 v1.1 -Page no. 10
Purpose of Data	Calculation of baseline emissions
Comments	The value is fixed ex-ante

For AMS III BG

Data / Parameter	CF
Data unit	-
Description	Default wood to charcoal conversion factor
Equations	Equation 1 (in AMS-III.BG)
Source of data	Default Value provided in Tool 33 version 2.0
Value applied	4
Justification of choice of data or description of measurement	A factor of 4 kg of firewood (wet basis) per kg of charcoal (dry basis).

methods and procedures applied	
Purpose of Data	Calculation of ER
Comments	This value is fixed for the crediting period Value is taken as default value mentioned in tool 33: Default values for Common Parameters

Data / Parameter	NCV_{wood}
Data unit	TJ/t
Description	Net Calorific Value of wood
Equations	Equation 1 (in AMS-III.BG)
Source of data	IPCC Default value mentioned on page number 10 of VMR0006 ver 1.1 methodology
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC Default Value
Purpose of Data	Calculation of Emission reductions
Comments	This parameter shall remain fixed for the crediting period

Data / Parameter	$NCV_{charcoal,default}$
Data unit	TJ/t
Description	Default net calorific value of charcoal
Equations	Equation 1 (in AMS-III.BG)
Source of data	
Value applied	0.0295
Justification of choice of data or description of measurement methods and procedures applied	The charcoal is produced using purely woody sources of biomass

Purpose of Data	Calculation of ER
Comments	Value is taken from IPCC 2006, Volume 2, Table 1.2 as described in the appendix 1 of the AMS-III.BG methodology. This value shall remain fixed for the crediting periods.

Data / Parameter	$EF_{projected\_fossilfuel}$
Data unit	t CO2/TJ
Description	Emission factor for the substitution of non-renewable woody biomass by similar consumers
Equations	Equation 1 (in AMS-III.BG)
Source of data	The value is taken as per default value mentioned in the applied methodology AMS II BG version 04.0, page number 10
Value applied	81.6
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of ER
Comments	This parameter shall remain fix for the crediting period

Data / Parameter	$GWP_{CH4,y}$
Data unit	t CO_2e/t CH_4
Description	Global warming potential of methane applicable to the crediting period
Equations	Equation 1 (in AMS-III.BG version 04.0)
Source of data	-
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	IPCC Default value
Purpose of Data	Calculation of emission reductions
Comments	this value shall remain fix for the crediting period

Data / Parameter	M_d
Data unit	tonne of CH4/tonne of raw material

Description	Factor to account for any legal requirement for capture and destruction of methane in the charcoal production facility. This parameter is equal to zero for the case of charcoal produced from micro-gasifier
Equations	Equation 1 (in AMS-III.BG version 04.0)
Source of data	The value of this parameter is sourced from the applied methodology AMS III BH version 04.0, page number 11
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Described as per the methodology
Purpose of Data	Equation 1 (in AMS-III.BG version 04.0)
Comments	This value remains fixed for the crediting period

6.2 Data and Parameters Monitored

For VMR0006

Data / Parameter	$N_{y,j,k}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for project activities and programme of activities. Alternately, simplified approach proposed in option (b) under Section 6.3 above may be used for determining the minimum sample size in which case compliance with 90/10 confidence precision is not obligatory.
Frequency of monitoring/recording	Every year
Value applied:	550,000
Monitoring equipment	Monitoring survey
QA/QC procedures applied	Determined using the monitoring survey. And also the Field Assistant visit each of the household once a month for the collection of the

	charcoal generated from the usage of the TLUD during the month. For a month on an average 29-30 kg of charcoal is generated using the micro-gasifiers, if the collected charcoal is more or less, than the field assistant will directly report to the manager for the same. The daily collection of charcoal can be verified by the daily collection report and the sales of charcoal (this happens weekly)
Purpose of data	For calculation of emission reduction
Calculation method	Not Applicable
Comments	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.

Data / Parameter	$\eta_{new,i,j}$
Data unit	%
Description	The efficiency of the device of each type i and batch j implemented as part of the project activity
Source of data	Calculated using the Equation no. 05 in methodology VMR0006 ver 1.1
Description of measurement methods and procedures applied	Adjustment of $\eta_{new,i,j}$ due to efficiency loss will be monitored according to VMR0006 version 1.1. Equation (5) "Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where the loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"
Frequency of monitoring/recording	Annually
Value applied:	26.32%
Monitoring equipment	Not Applicable
QA/QC procedures applied	Not Applicable
Purpose of data	Calculation of emission reductions
Calculation method	As per equation 5 in Methodology VMR0006 ver 1.1 $\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$

	<table><tr><th>Year</th><th>Efficiency</th></tr><tr><td>0</td><td>26.32%</td></tr><tr><td>1</td><td>26.06%</td></tr><tr><td>2</td><td>25.80%</td></tr><tr><td>3</td><td>25.54%</td></tr><tr><td>4</td><td>25.28%</td></tr><tr><td>5</td><td>25.03%</td></tr><tr><td>6</td><td>24.78%</td></tr></table>
Comments	

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	*Tonnes*
Description	Annual quantity of woody biomass used by improved Cookstoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey
Source of data	Monitoring Survey: (Survey on $fillings_{y=1}$ and $load_{y=1}$ as described below.)
Description of measurement methods and procedures applied	$B_{y=1,new,survey}$ = $fillings_{y=1}$ * $load_{y=1}$ * 0.052 (conversion factor to from kg/week to t/a) $fillings_{y=1}$: Average number of weekly fillings of a batch-loaded micro gasifier stove $fillings_{y=1}$: 31.13 A representative sample of randomly selected users was asked for the average number of weekly loads of fuelwood burnt in their micro gasifier stove, specifying loads per specific meal and days of usage per week. Where micro gasifier stove are found not to be operational during monitoring, they will not considered here. $load_{y=1}$, : Average amount of fuelwood used per filling of a micro gasifier stove $load_{y=1}$, : 1.088 kg The average weight of a micro gasifier stove fuelwood load was determined as the average value obtained over all monitoring survey conducted for determining the average woodfuel consumption. $B_{y=1,new,survey}$ = $fillings_{y=1}$ * $load_{y=1}$ * 0.052 (a) *$B_{y=1,new,survey}$ = 31.13 *1.088kg *0.052 = 1.76*

Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied:	1.76
Monitoring equipment	Monitoring Survey
QA/QC procedures applied	Not Applicable
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable
Comments	Survey would be conducted in first year of implementation to estimate the baseline wood consumption in the project device.

Data / Parameter	Life span
Data unit	*Years*
Description	Project promoters to state the operating lifetime of project device for projects opting Equation 5 for determining project stove efficiency.
Source of data	Manufacturer's specifications
Description of measurement methods and procedures applied	It is taken from the manufacturing specification of the cookstove.
Frequency of monitoring/recording	Once at the time of Project stove installation
Value applied:	7 years
Monitoring equipment	Not required
QA/QC procedures applied	Not Applicable
Purpose of data	Not Applicable
Calculation method	Not Applicable
Comments	This remain fixed for the life cycle of the cookstove

Data / Parameter	$N_{y,i}$
Data unit	Number
Description	Number of project devices of type i operating during year y
Source of data	Monitoring and surveys
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for project activities and programme of activities Alternately, simplified approach proposed in option (b) under Section 8.4 above may be used for determining the minimum sample size in which case compliance with 90/10 confidence precision is not obligatory.
Frequency of monitoring/recording	Annual
Value applied:	550,000
Monitoring equipment	Not Applicable
QA/QC procedures applied	Not Applicable
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable
Comments	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value

For AMS-III.BG

Data / Parameter	$Q_{CCP,i,y}$
Data unit	Tonnes
Description	Produced quantity of charcoal product i in year y
Source of data	Charcoal FA(field Assistant) Daily report.

	charcoal sales bill issued by eligible retailer.
Description of measurement methods and procedures applied	The parameter can be monitored according to one of the following options: Option1: Direct measurement (e.g. use of a scale) of the weight of charcoal products supplied; Option 2: Calculation of the total weight of charcoal supplied; based on the total number of bags supplied and the average weight of charcoal product per bag. The weight of charcoal products per bag is determined on sample basis in accordance with the sampling standard (e.g. using systematic sampling method). Option 2 can only be used if Option 1 is not available. Charcoal generated from only woody biomass is eligible for the case of micro-gasifier Option 1 is chosen for the determination of the parameter and $Q_{CCP,i,y}$ will be derived from invoices/receipts of sales of charcoal generated in the project to charcoal users and retailers. If feasible, the weight of charcoal delivered will be indicated on invoices, requiring the availability of calibrated weighbridges or other scales. Where invoicing to charcoal buyers cannot based on the exact weight but on the number of bags delivered, the following formula will apply: $Q_{CCP,i,y}=weightbags_{y,i} * n_{bags,y,i}$ Where: $weightbags_{y,i}$ average weight of bags of charcoal type i in period y $n_{bagsy,i}$ total number of bags of charcoal type i in period y $n_{bagsy,i}$ will be determined based on invoices/receipts of charcoal buyers. $weightbags_{y,i,}$ will be determined on sample basis in accordance with the sampling standard. Simultaneously, it will be checked that charcoal buyers are eligible according to para. 4 of AMS-III.BG version 04. The by-product charcoal generated from operation of TLUD is collected by charcoal Field Assistant *(FA)* assigned for collection of specific range of EK-TLUD. The collection of charcoal happens once a month from a household, i.e., once a month the FA would come at house for the collection of the charcoal. Charcoal FA will check the condition of charcoal, weighing of charcoal using a non-calibrated hand held electronic weighing scale, and payment is made to the user at fixed priced per kg of charcoal. The data is entered into the FA daily Sheet

	which then becomes the master document for charcoal collection reporting. And collected charcoal is stored at FA place. Weekly basis, the contracted charcoal retailer comes to a central point and picks up the charcoal. The FAs brings the charcoal from their home to the pickup point. The charcoal buyer weighs charcoal bags of 25 kg thru random sampling and loads in their trucks. A charcoal sales bill or invoice is generated for receiving payment from the vendor and the same invoice becomes the master document for charcoal sales calculations. The bill template with the contracted retailer clearly states the weight of charcoal, number of standard(25kg) bags, and payment amount.
Frequency of monitoring/recording	Continuously at the time of collection from TLUD end-user and delivery to contracted charcoal buyers or retailers Weekly updated to database
Value applied:	Will be determined in the monitoring periods
Monitoring equipment	Calibrated weighing scale and standard sized bags for charcoal.
QA/QC procedures applied	The procedure and eligibility criteria is determined in TLUD GASIFIER CARBON CREDIT PROJECT – FIELD IMPLEMENTATION PROCESS NOTE agreed between ENCC and Sapient.
Purpose of data	Calculation of emission reduction
Calculation method	The entire chain of charcoal collection will be documented, demonstrating how the amount of charcoal delivered to users of conventional charcoal relates to the amount generated by users. There will be cross checks with: - total quantity of charcoal generated by the micro gasifiers based on monitored fuelwood consumption and the conversion-rate to charcoal, which is 4. - the average amounts of charcoal collected from stove users based on records of field assistant collecting charcoal
Comments	Charcoal under this project is sold in Standardized bags of 25kg each which contain the 25 kg of charcoal. Bags are weighed at random at the point of sale in a calibrated weighing scale

Data / Parameter	$NCV_{Charcoal,i}$
Data unit	TJ/tonne

Description	Net calorific value of the charcoal i produced
Source of data	Option II of the Approved methodology.
Description of measurement methods and procedures applied	The value can be determined according to one of the following options: Option 2: Using one of the options provided in the appendix of the methodology AMS III BG version 04.0; Option 1: For charcoal from coconut husks, bamboo and other purely wood sources of biomass, the following assumptions could be made: $NCV_{charcoal,I}$ = 29.5 GJ/tonne i.e. $NCV_{charcoal,i}$ = 0.0295 TJ/tonne
Frequency of monitoring/recording	Parameter will be monitored at each monitoring period.
Value applied:	0.0295
Monitoring equipment	-
QA/QC procedures applied	The fuelwood is used in the micro-gasifier which produces the charcoal as a by-product of cooking.
Purpose of data	Calculation of emission reductions
Calculation method	
Comments	

6.3 Monitoring Plan

The purpose of the monitoring plan is to ensure successful monitoring of the emission reductions of the proposed project during its crediting period. The overall monitoring will be managed by the project implementing partner Sapient InfoTech, India

The information in these databases is updated continuously, whenever new data (distribution contracts) are available. Original copies of the distribution contracts will be kept and maintained for two years after the end of the crediting period.

As for charcoal sales, there is a database on charcoal sales, based on invoices/receipts of sales to charcoal buyers.

The monitoring activities will involve data collection during distribution as well as usage information post distribution. The data collected during distribution will involve information about the stove, the purchaser and location to enable one to uniquely identify each TLUD unit and avoid double counting. This will form what is called the sales record. Sapient InfoTech will distribute the stoves, as well as carry out the monitoring activities that occur during the distribution of stoves. Sapient InfoTech or through their associates will ensure that the distributor is trained on how to capture the sales data. The project activity implementer is fully responsible to ensure the correct distribution process and data gathering is followed. In turn, the distributing agent will link remuneration to the complete and accurate collecting of information during the distribution of stoves.

The following information has been recorded at the time of distribution of the micro gasifiers and ICS to the user:

- Stove ID
- Date of stove distribution
- Field Assistant
- Beneficiary Details
    - Name of user
    - Spouse Name
    - Village
    - Post Office & Police Station
    - District
    - GPS
- Scans of the distribution contracts
- Other relevant information in accordance with relevant GHG program

However, during the post distribution, monitoring activities will involve selecting a sample of stoves from the sales record and visiting the premises where these stoves are located to monitor the key parameters pertinent in ER calculations.

Data collecting & handling is conducted in a transparent way to secure high quality of recording and storing of data. Data collected and monitored are stored electronically in a secure and retrievable manner for at least two years after the end of the project crediting period. Uncertainty related to data handling (if any) would be rectified, if necessary by revising monitoring procedures. The changes would be approved by a verifier (e.g. Designated Operational Entity)

Sampling Plan:

As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04, the sampling plan is the following:

(a) Sampling Design

Due to the large number of stoves envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project instance-wide Sampling Plan (by grouping and sampling across project activities) that is designed in line with the requirements of the "Sampling and surveys for CDM project activities and programme of activities", version 04.

(i) **Objective and Reliability Requirements:**

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per CDM Methodology AMS-II. G, 90/10 confidence/precision shall be applied for the annual sampling requirement and 95/10 for biennial sampling inspection. As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, 90/10 confidence/precision to be adopted for the small-scale project and 95/10 for the large-scale project. Given that the size of the project is under the category of large scale projects, hence 95/10 confidence/precision shall be adopted for all parameters unless the average annual emission reductions of the project are below the threshold.

Monitored Parameter-

Parameter	Description	Parameter type/ Frequency
$N_{y,i,j}$	Number of project device of type i and batch j operating during year y	Proportion/ Biennially
$B_{y=1,new,i,j,survey}$	Quantity of woody biomass consumed in project device	Proportion/determined for the first time
$Q_{CCP,i,y}$	Produced quantity of charcoal produce in i in year y	Proportion/ determined in the first monitoring period
$NCV_{charcoal,i}$	Net calorific value of the charcoal produce	Proportion/ determined in the first monitoring period

Target Population

the target population will be the complete set of the appliances (stoves) deployed / beneficiaries using stoves under the project

Sampling method

The project involves distribution of micro-gasifier throughout the project area thereby replacing traditional cooking devices. The population is homogenous in nature i.e. common technology with similar operating characteristics. The characteristics of population (for example quantity of biomass consumed) are more similar within the population (Micro gasifier of same type, vintage and project area in which they are operating). Therefore, Simple Random Sampling technique will be used to conduct sampling survey among ICS batches. The populations of each batch will be combined together, the sample size is determined, and a single survey will be undertaken to collect data. To ensure the survey result is representative of the entire population, the dissimilarity (such as Stove type, vintage and project area in which they are operating) within the included Project activities will be taken into account in the sample size calculation. The stove of same type, vintage and project area in which they are operating will be grouped in the same strata. Samples will be drawn by using the random number generator.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$- = Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$B_{y=1,new,i,j,survey}$ = Interview with end user and estimate the daily consumption of woody biomass of stoves (Daily consumption of woody biomass

Sample Size

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 95 per cent confidence level, and a 10 per cent margin of error, a random sample will be selected from population. In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0, there are different ways available to obtain the estimates of the parameter of interest:

(a) Refer to the result of previous studies and use these results;

(b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;

(c) Use best guesses based on the researcher's own experiences.

Proportion parameter to estimate the sample size for proportion parameters, the following equation is used:

$$n \geq \frac{1.96^2 Np(1-p)}{(N-1) \times 0.1^2 + 1.96^2 p(1-p)}$$

Equation (8)

Where;

n	=	Sample size
N	=	Population size (total number of beneficiaries)

p = Weighted overall expected proportion

1.96 = Represents the 95% confidence required

0.1 = Represent the 10% relative precision

Sample size calculation

For the current monitoring period where only 2000, TLUD micro-gasifiers have been distributed the Sample size calculation can be done as

$$n \geq \frac{1.96^2 Np(1-p)}{(N-1) \times 0.1^2 + 1.96^2 p(1-p)}$$

Equation (9)

Where;

n = Sample size

N = Population size (total number of beneficiaries) (2,000)

p = Weighted overall expected proportion (0.90)

1.96 = Represents the 95% confidence required

0.1 = Represent the 10% relative precision

$$n \geq \frac{1.96^2 \times 2000 \times 0.9(1-0.9)}{(2000-1) \times 0.1^2 + 1.96^2 \times 0.9(1-0.9)}$$

$$n \geq 42$$

A sample size of 50 households were then selected using the random selection function of the excel. The selected households are

Number of sampled cookstoves									
1349	744	1633	649	1480	1660	1059	250	1259	558
1319	1374	167	1648	1081	949	1396	1793	613	1861
70	1860	1543	1686	925	1063	1846	107	312	1726
274	1961	601	1799	936	1946	1717	1223	174	26
1197	1976	636	1038	906	1825	626	305	5	299

The monitoring period for the first monitoring is from 10-September-2022 to 31-December-2022. And for which the monitoring of the parameters was carried out from 24-Februrary-2023 to 27-February-2023. A simple random sampling was done to determine the sample size for the monitoring of the households and a total of 50 samples were surveyed. The selection of sample households was on the random basis, using the sampling technique in the excel worksheet.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

The monitoring period for the first monitoring is from 10-September-2022 to 31-December-2022. And for which the monitoring of the parameters was carried out from 24-Februrary-2023 to 27-February-2023. A simple random sampling was done to determine the sample size for the monitoring of the households and a total of 50 samples were surveyed. The selection of sample households was on the random basis, using the sampling technique in the excel worksheet.

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,j,k}$
Data unit	*Number*
Description	Number of project devices of type i and batch j operating during year y
Value applied:	2,000
Comments	During this monitoring period only 2000 TLUD micro-gasifiers were distributed.

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Quantity of woody biomass used by the project devices in tonnes per device of type i
Value applied:	0.4679
Comments	Determined by monitoring survey

Data / Parameter	$Q_{CCP,i,y}$
Data unit	Tonnes

Description	Produced quantity of charcoal product i in year y
Value applied:	213.25
Comments	This is total charcoal which is produced and sold. This can be verified by the invoices.

Data / Parameter	$NCV_{Charcoal,i}$
Data unit	TJ/tonne
Description	Net calorific value of the charcoal i produced
Value applied:	0.0295
Comments	As per the Option II in the applied methodology AMS II BG

7.2 Baseline Emissions

The methodologies VMR0006 and AMS III BG does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

7.3 Project Emissions

The methodology VMR0006, does not account for project emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. Please refer to Section 4.4

The methodology AMS III BG, the project emission during the use of micro-gasifier cookstove is taken as zero.

7.4 Leakage

Leakage has considered as a default value of 0.95 in accordance with Section 5.4 of AMS II G

A default value of 0.030 t CH4/t charcoal may be used in accordance with “AMS-III.BG.: Emission reduction through sustainable charcoal production and consumption

Emission reduction calculations for this monitoring period is as follows:

From VMR0006 version 1.1, the due to savings of the fuelwood consumption by the use of micro-gasifier.

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times \left(EF_{wf,CO2} + EF_{wf,non\ CO2}\right) \times N_{y,i,j} \times 0.95$$

Where;

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y (0.4679 *tonnes/monitoring period)*
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) (84.48%)
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)
$EF_{wf,CO2}$	=	CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO2/TJ)
$EF_{wf,non\ CO2}$	=	Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO2/TJ)
$N_{y,i,j}$	=	Number of improved cook stoves of type i and batch j operating during year y (2,000)
0.95	=	Discount factor to account for leakage

$$ER_{y,i,j} = 0.4679 \times 84.48\% \times 0.0156 \times (112 + 26.23) \times 2000 \times 0.95$$

$$\boldsymbol{ER_{y,i,j}} = \mathbf{2,643}\ \textbf{tonnes CO}_2\textbf{e/monitoring period}$$

From AMS III BG; version 04.0; for the sustainable charcoal production by the use of the micro-gasifier

$$\begin{aligned} ER_y = \sum_i Q_{CCP,iy} & \\ & \times \left[\left(CF \times NCV_{wood} \times \frac{NCV_{charcoal,i}}{NCV_{charcoal,default}} \times f_{NRB,BL,wood} \times EF_{projected\_fossilfuel}\right)\right. \\ & \left. + (SMG_{y,b} - M_d) \times (1 - f_{NRB}) \times GWP_{CH4,y}\right] - PE_{y,fugitive} - PE_{y,flaring} \\ & - PE_{FF,y} - PE_{EL,y} - PE_{BC,y} \end{aligned}$$

$Where;$

ER_y	=	Emission reductions in year y (t CO2e/yr)

$Q_{CCP,iy}$	=	Quantity of charcoal type i produced and used in year y (t)
CF	=	Wood-to-charcoal conversion factor
NCV_{wood}	=	Net calorific value of wood (TJ/t)
$NCV_{charcoal,i}$	=	Net calorific value of the charcoal type i produced during the project (TJ/t)
$NCV_{charcoal,default}$	=	Default net calorific value of charcoal (TJ/t)
$f_{NRB,BL,wood}$	=	Fraction of biomass used in the absence of the project activity that can be established as non-renewable biomass (fraction or %)
$EF_{projected\_fossilfuel}$	=	Emission factor for the substitution of non-renewable woody biomass by similar consumers (t CO2/TJ)
$GWP_{CH4,y}$	=	Global warming potential of methane applicable to the crediting period (t CO2e/t CH4)
$SMG_{y,b}$	=	Specific methane generation for the baseline charcoal generation process in the year y (tonnes CH4/t charcoal product); a default value of 0.030 t CH4/t charcoal may be used. Alternatively, the value can be determined in accordance with the procedure provided in the latest version of AMS-III.K.
M_d	=	Factor to account for any legal requirement for capture and destruction of methane in the charcoal production facility (tonne of CH4/tonne of raw material) This is taken as Zero as there is no legal requirement for capturing of the methane.
$PE_{y,flaring}$	=	If applicable, emissions due to the flare inefficiency in the project charcoal manufacturing plant in the year y (t CO2e) determined in accordance with the procedure provided in AMS-III.K. In case captured pyrolysis gas is gainfully used (e.g., as fuel for pre-heating the facility, or for wood drying, or used for production of heat and/or power as in the case of micro-gasifier), then it can be taken as zero This is taken as Zero
$PE_{FF,y}$	=	Project emissions due to fossil fuel consumption in charcoal production facilities in year y (t CO2)

		Since, charcoal production is a by-product to the micro-gasifier therefore taken as zero.
$PE_{EL,y}$	=	Project emissions due to electricity consumption in charcoal production facilities in year y (t CO2) Since, charcoal production is a by-product to the micro-gasifier therefore taken as zero.
$PE_{BC,y}$	=	Project emissions due to biomass cultivation in year y (t CO2) There is use of the fuelwood which is taken from the field and buyed, this fuelwood is then used in the micro gasifier and the by-product of the combustion process is micro-gasifier therefore, taken as zero.

Including the project emissions due to fugitive emissions:

$$PE_{y,fugitive} = \sum_i Q_{CCP,i,y} \times GWP_{CH4,y} \times SMG_{y,b} \times f$$

$PE_{y,fugitive}$ = Fugitive emissions from operation of charcoal producing facility(physical leakage) in the year y (t CO_2e)

f = A fraction attributed to project charcoal production technology, use a default value of 0.1.

Combining the above two equations

Which is transformed to:

$$ER_y = \sum_i Q_{CCP,iy} \times \left[\left(CF \times NCV_{wood} \times \frac{NCV_{charcoal}}{NCV_{charcoal,default}} \times f_{NRB,BL,wood} \times EF_{projected\_fossilfuel}\right) + SMG_{y,b} \times (0.9 - f_{NRB}) \times GWP_{CH4,y}\right]$$

$$ER_y = 213.45 \times \left[\left(4.0 \times 0.0156 \times \frac{0.0295}{0.0295} \times 84.48\% \times 81.6\right) + 0.03 \times (0.9 - 84.48\%) \times 28.0\right]$$

$$ER_y = 213.25 \times \left[\left(4.0 \times 0.0156 \times \frac{0.0295}{0.0295} \times 84.48\% \times 81.6\right) + 0.03 \times (0.9 - 84.48\%) \times 28.0\right]$$

$$ER_y = 927.22\ tonne\ \ CO_2e/monitoring\ period$$

Accounting the leakage of 0.05% to the calculations

$$ER_y = 927.22 \times 0.95 \, tonne \;\; CO_2e/monitoring \; period$$

$$\boldsymbol{ER_y = 880 \, tonne \;\; CO_2e/monitoring \; period}$$

Total Emission reduction in this monitoring period starting from 10 September 2022 to 31 December 2022 is

$$\boldsymbol{ER_y = ER_{VMR0006,y} + ER_{AMS\,II\,BG,y}}$$

$$\boldsymbol{ER_y = 2,643 + 880}$$

$$\boldsymbol{ER_y = 3,444 \, tonne \;\; CO_2e/monitoring \; period}$$

7.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
Year 2022 10-Sept-2022 to 31-Dec-2022	3,444	0	0	3,444
Total	3,444	0	0	3,444

Ex-ante emission reductions	Achieved Emission Reductions	Percentage Difference	Justification for the Difference
3,951	3,444	-12.83%	Distribution of the cookstove was only 2000 was carried out during the monitoring period. This distribution was carried out on different dates starting from 10 September 2022 to 27 September 2022.

APPENDIX I: BASELINE DATA FORM

TLUD – BASELINE DATA

NAME: ____

ADDRESS: ____

____

____

GPS COORDINATES: ____

1. **Which cooking device are you using for your household cooking currently?**

   Mud Stove(wood) : 1
   Improved Wood Cook Stove : 2
   LPG : 3
   Kerosene Stove : 4
   Other(Specify) : 5

2. **What do you use as cooking fuel currently? In case of multiple fuel what is the proportion of each type? What quantity of each type per month? Rate at which you buy per kg/litre?**

Fuel Type		Proportion	KG/litre p.m	Rate	Cost/Month
Fire Wood	1				
Cow Dung Cakes	2				
Kerosene Oil	3				
LPG	4				
Others(Specify)	5				

3. **How do you procure the fuel that you use for cooking?**

   Buy it : 1 Collection : 2

   In case if both buy & collect: What proportion is bought & what proportion is collected?

   Bought: ______% Collection _________%

4. **What kind of food do you prepare in your household? What is the approx. Time taken to cook each type of dish?**

DISHES	TIME TAKEN TO COOK

5. **How many time do you cook in a day:**

   Once : 1 Twice : 2 > Twice : 3

6. **How many family members do u have?**

   Ages 0-10 : ______
   11-18 : ______
   18-60 : ______
   60+ : _______

7. **How many mud-cookstoves already present in the house?**

   ____

APPENDIX II: SAMPLE OF END USER AGREEMENT

End User Agreement between CME and Beneficiary

This agreement is made between Eco Nexus Carbon Credits, The Coordination and Managing Entity (CME) of the Carbon Credit Project **'Sustainable charcoal and Improved cook stoves initiative using micro gasifier in India'** having its registered office at , **369, Gangnam-daero, Seocho-gu, Seoul, Republic of Korea (hereinafter referred to as "ENCC")** on behalf of Project Proponent **AND** The beneficiary regarding the purchase of a TLUD stove. ENCC as Project Owner & CME together with Sapient Infotech, Kolkata is selling the TLUD Stove **at a highly subsidized price of INR 1250**

Stove ID	Date of stove distribution	Field Assistant	Beneficiary Details					
			Name of user	Husband Name	Village	Post Office & Police Station	District	GPS

- The beneficiary confirms receipt of a TLUD stove and confirms that up to now wood has been used as cooking fuel in a traditional stove.
- Beneficiary will participate in monitoring and verification by CME – ENCC and Sapient Infotech. The resulting data are used and evaluated only within a carbon credit project. The beneficiary is aware that he/she receives the TLUD stove at a highly reduced price and therefore cede all entitlements resulting from reduced greenhouse gas emissions to ENCC, Seoul, Republic of Korea.
- Beneficiary will not sell the stove. If the stove is transferred to another address or to another owner or the stove is stolen / not used anymore, beneficiary shall inform the relevant field assistant immediately.

Signature Field Assistant (on behalf of CME): Signature/Thumb Impression of Beneficiary:

____________________________ ____________________________

APPENDIX III: MONITORING SURVEY FORM

Monitoring Survey

-questionnaire user interviews-

General information:	
Project:	**VCS 3954**
Monitoring Period:	MP 1 Please put relevant dates
Date of interview:	**(day)** **(month)** **(year)**
Name of interviewer:	
Type of contact	□ On-Site Visit □ Telephone Interview

User and stove information:		
Stove-ID: ___ ___ ___ ___	Name of person interviewed:	Relation to stove user who signed the contract: ___ ___ ___ _____ ___ ___ ___

Contact details of user correct (compare with database entry)?		If contact details not correct, please correct here: ___ ___ ___ _____ ___ ___ ___
Yes	**No**	
□	□	

Questions to the user:			
1) Is the TLUD in use?		**Yes** □	**No** □
If not in use, why?	___ ___ ___ ___ ___ ___ ___ _____ ___ ___		
2) What price did you pay for the TLUD?	___ ___ ___ ___ **INR**		
3) On how many days per week do you use the TLUD?	1_ 2_ 3_ 4_ 5_ 6_ 7_		
4) Please describe how many times you fill the canister in each cooking session on a typical day of usage.	Breakfast __ times Lunch __ times Diner __ times **Entire day** __ times **Comments** ___ ___ ___ ___		
5) What do you do with the charcoal generated in the TLUD?	**Sold to field assistant** □	**Sold to other (please specify)** ___ ___ ___ ___	**Not sold** □

6) Do you know how many kg of charcoal you normally generate in a month?	__ __ __ __ kg			
7) Compared to the traditional stove you used, how much smoke does the TLUD generate?	**Much Less**	**Less**	**Equal**	**More**
	◻	◻	◻	◻
8) Do you use LPG for cooking?	**Yes**		**No**	
	◻		◻	
9) What kind of stove did you use before using the TLUD?	__ __ __ ___ __ __ __			
10) Has the pre-project device completely been decomissioned after you received the TLUD?	**Yes**		**No**	
	◻		◻	
11) Is the TLUD the only stove you are using for cooking? If no, please specify below	**Yes**		**No**	
	◻		◻	
a. On how many days per week do you use the other stoves for cooking?	1_ 2_ 3_ 4_ 5_ 6_ 7_			
b. How many times per day do you use the other stoves for cooking?	Breakfast __ times Lunch __ times Diner __ times **Entire day** __ times **Comments** __ __ __ __			
12) Fuel consumtion a. Wood consumption in a batch:				
b. Is the wood consumtion is decreased in TLUD as compared to pre-project device?	Yes ( )		No ( )	
c. Fuelwood collection time in TLUD as compared to pre-project device?	Increased ( )		Decreased ( )	
d. Cooking time in TLUD as compared to pre-project device?	Increased ( )		Decreased ( )	
13) Any additional comments?	__ __ __ ___ __ __ __			

Signature of **Respondent**: __ __ __ ___ __ __

Signature of **Interviewer**: __ __ ___ __ __ __

APPENDIX IV: TECHNICAL SPECIFICATION OF TLUD MICRO-GASIFIER

SAPIENT INFOTECH

Dated: 5th October 2022

ANNEX: EK TLUD

TO WHOMSOEVER IT MAY CONCERN

The specifications of the Champion EK TLUD stove manufactured by Sapient Infotech, Kolkata for Eco Nexus Carbon Credits West Bengal, India, project are given below:

Stove firepower	5.471	kW, based on WBT test results CPA1 MP1
Stove efficiency	28	% based on MSME WBT Certificate
Capacity per TLUD	**1.516**	**kWth**

Sapient Infotech hereby certifies that the life of the stove will be 7 years. A 2 years unconditional warranty will be provided for each stove

Sapient Infotech confirms that the TLUD stoves manufactured under the Eco Nexus Carbon Credit project in West Bengal, India, will have unique running numeric ID starting from 1 with prefix of EK.

Sapient Infotech further confirms that for the ENCC West Bengal, India, project, neither Sapient Infotech nor any of their project partners other than Eco Nexus Carbon Credits involved in project activities will claim any Carbon Credit from the stoves manufactured & distributed under the EK prefix unique ID

For Sapient Infotech

Proprietor

Champion TLUD Specifications :

Sr. No	Parts	Material of construction (MOC)	Size in MM
1	OUTER	SS 304	185X305X0.5
2	INNER	SS 304	150X205X0.5
3	BOTTOM	SS 202	150X100X0.5
4	SLEEVE	SS 202	143X175X0.5
5	L CLAMP	M S YELLOW PLATED	75X75X2.5
6	BIG HANDLE	PP(GF)	265X35
7	SMALL HANDLE	PP(GF)	181X25
8	MESH	SS 202 (Spiral)	148X3
9	TRIPOD	M S YELOW PLATED	405X16
10	TAWA	CAST IRON POWDER COATED	278X3.5
11	CONNECTOR	SS 202	128/118X45X.5
12	VESSEL SUPPORT	M S POWDER COATED	190X190X1
13	TOP PLATE	SS 304	206X26X0.5
14	PLUNGER	M S YELLOW PLATED	280X8X1
15	CONNECTOR ROD	PP(GF)	140X8X1
16	CONNECTOR BUSH	PP(GF)	13X26X4

GROSS WEIGHT : 8 kg